Confidence Calibration under Ambiguous Ground Truth

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Abstract—Confidence calibration assumes a unique ground-truth label per input, yet this assumption fails wherever annotators genuinely disagree. Post-hoc calibrators fitted on majority-voted labels can appear well-calibrated under voted-label evaluation yet remain substantially miscalibrated against the underlying annotator distribution. We show that this failure is structural: under simplifying assumptions, Temperature Scaling is biased toward temperatures that underestimate annotator uncertainty, with true-label miscalibration increasing monotonically with annotation entropy. To address this, we develop a family of ambiguity-aware post-hoc calibrators that optimise proper scoring rules against the full label distribution and require no model retraining. Our methods span progressively weaker annotation requirements: **Dirichlet-Soft** leverages the full annotator distribution and achieves the best or tied-best Brier score and NLL in 7 of 8 settings; **Monte Carlo Temperature Scaling** with a single annotation per example (MCTS $S{=}1$) matches full-distribution calibration within 0.6 percentage points ECE across all benchmarks, demonstrating that pre-aggregated label distributions are unnecessary; and **Label-Smooth Temperature Scaling (LS-TS)** operates with voted labels alone by constructing data-driven pseudo-soft targets from the model's own confidence. Experiments on four benchmarks with real multi-annotator distributions (CIFAR-10H, ChaosNLI) and clinically-informed synthetic annotations (ISIC 2019, DermaMNIST) show that Dirichlet-Soft reduces true-label ECE by 59–86% relative to Temperature Scaling, while LS-TS reduces ECE by 7–78% without any annotator data.

Index Terms—Confidence calibration, ambiguous ground truth, annotator disagreement, temperature scaling, soft labels, proper scoring rules, label distribution learning.

1 Introduction

Deep neural networks are increasingly deployed as decision-support tools in high-stakes domains. In medical imaging, classifiers assist dermatologists in diagnosing skin lesions and pathologists in identifying malignant tissue [1], [2]; in natural language processing, models flag toxic content, assess clinical notes, and support legal document review; in autonomous driving, perception systems make real-time safety decisions under uncertainty [3]. In each of these settings the model outputs not merely a predicted class but an implicit *reliability claim*: the confidence score attached to a prediction is understood by downstream systems and human operators as a direct probability that the prediction is correct. A radiologist who sees 90% confidence on a malignant-lesion classification may reduce the level of follow-up scrutiny; a content moderator acting on a 99% toxicity score may take irreversible action; an autonomous vehicle assigns braking decisions based on its detection confidence. The faithfulness of stated confidence to empirical correctness — calibration — is therefore as important to deployment safety as accuracy itself [4], [5].

Confidence calibration. This property is formalised as follows. A well-calibrated classifier $f : \mathcal{X} \to \Delta^K$ produces probability vectors $\hat{p}(x) = f(x)$ such that

$$\mathbb{P}(\hat{Y} = Y \mid \hat{p}_{\hat{Y}}(X) = p) = p, \quad \forall\, p \in [0, 1], \tag{1}$$

where $\hat{Y} = \arg\max_k \hat{p}_k(X)$ and Δ^K denotes the K-dimensional probability simplex. Guo et al. [6] showed that modern high-accuracy networks are systematically overconfident, and proposed Temperature Scaling (TS) — dividing logits by a scalar $T > 1$ before softmax — as an effective post-hoc remedy. Subsequent work extended post-hoc calibration to Platt scaling [7], isotonic regression [8], and Dirichlet calibration [9], and studied its behaviour across architectures [10] and tasks [11]. These methods share a common design principle: they fit a correction to the model's logits using a held-out calibration set annotated with *one-hot labels*.

The hidden assumption. The calibration condition (1) implicitly assumes that every input x has a unique ground-truth label Y. This assumption is violated whenever the task is inherently ambiguous or subjective. A skin lesion may be legitimately classified as malignant or benign by different dermatologists, depending on subtle visual features that experts genuinely disagree about [1]; a natural-language premise may or may not entail a hypothesis depending on pragmatic interpretation [12]; a low-resolution object may be categorically ambiguous even to careful human observers [13]. Such disagreement is not annotation *noise* to be corrected — it is irreducible *aleatoric* uncertainty inherent to the task. The correct calibration target is therefore not a single label Y but a *distribution over labels* $\pi(\cdot \mid x) \in \Delta^K$, which we call the *annotator distribution*, representing the probability that a randomly selected expert would assign each label to input x. Figure 1 illustrates three concrete instances of this phenomenon across image classification, natural language inference, and medical imaging.

The standard practice and its failure. Standard practice aggregates annotations by majority vote to obtain a single *voted label* y^* and calibrates against it — and in many practical

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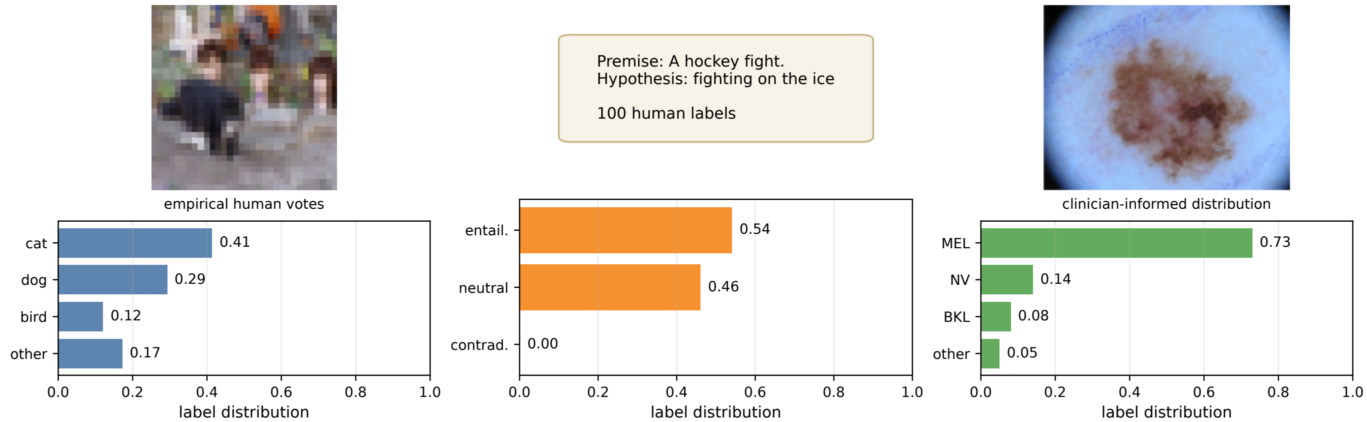


Fig. 1: **Left:** a CIFAR-10H image with dispersed human votes (cat/dog/bird), illustrating perceptual ambiguity in low-resolution vision. **Middle:** a ChaosNLI premise–hypothesis pair with split entailment/neutral judgments, illustrating semantic ambiguity. **Right:** an ISIC 2019 melanoma image paired with the clinician-informed label distribution used in our medical experiments, showing clinically plausible MEL/NV confusion. CIFAR-10H and ChaosNLI distributions are empirical human label distributions; the ISIC panel uses the dermatologist confusion model described in Appendix I.

settings this one-hot label is the only supervision available. This paper demonstrates that this is not merely a rough approximation but a *systematic failure*: calibrating to the voted label optimises the wrong objective, and the resulting miscalibration worsens as genuine annotator disagreement increases. A particularly revealing observation is that the most expressive voted-label calibrator (Dirichlet calibration with a full $K\times K$ affine transform) consistently *underperforms* simple Temperature Scaling in true-label ECE — additional flexibility with the wrong target exacerbates rather than alleviates the problem. In a controlled experiment (Section 4), every standard voted-label calibrator fails to close the gap; the root cause is the calibration *target*, not the method's capacity. This observation parallels the finding of Stutz et al. [14] for conformal prediction, where sets calibrated on voted labels systematically undercover the true annotator distribution.

Contributions.

1) **Problem and theory** (Sections 3–5): we formalise true-label calibration under ambiguous ground truth and prove that Temperature Scaling is biased toward unduly low temperatures, with the miscalibration gap growing monotonically with annotation entropy.
2) **Methods** (Section 6): we propose a family of ambiguity-aware post-hoc calibrators that use the annotator distribution as the calibration target and require no model retraining, including: **Dirichlet-Soft** (full distribution available); **MCTS** (individual annotations available — a single annotation per example, $S{=}1$, already matches full-distribution calibration); and **LS-TS**, an annotation-free variant that uses only voted labels yet substantially closes the calibration gap.
3) **Evidence** (Sections 4 and 7): experiments across four benchmarks spanning vision (CIFAR-10H, ISIC 2019, DermaMNIST) and language (ChaosNLI) show that Dirichlet-Soft reduces $\mathrm{ECE}_{\mathrm{true}}$ by 59–86% and LS-TS by 7–78% relative to TS, while standard calibrators can actively worsen true-label ECE.

2 Related Work

Confidence calibration. Guo et al. [6] established that modern high-accuracy networks are systematically overconfident and that Temperature Scaling (TS) is a simple and effective post-hoc remedy. The multiclass calibration literature has since proposed isotonic regression [8], Platt scaling [7], Dirichlet calibration [9], and adaptive per-instance temperature variants [15]. Minderer et al. [10] showed that Vision Transformers are inherently better calibrated than CNNs, and Bai et al. [11] revisit when TS is and is not sufficient. Kumar et al. [16] propose verified uncertainty calibration through non-parametric testing. All of these methods share a common assumption: the calibration set contains one-hot labels, and calibration means aligning confidence with correctness relative to those labels. We show that this assumption — not the method's architectural capacity — is the limiting factor when labels are genuinely distributional.

Soft labels, label smoothing, and knowledge distillation. Training-time approaches have explored the benefits of soft targets. Label smoothing [17], [18] adds a fixed uniform component to one-hot targets and has been observed to improve both accuracy and calibration; however, it applies a dataset-global constant regardless of per-instance annotator disagreement. Knowledge distillation [19] uses teacher model outputs as soft training targets, improving student generalisation and calibration. Thulasidasan et al. [20] show that Mixup training substantially improves calibration by encouraging confident predictions only when training examples are well-separated. Collins et al. [21] collect CIFAR-10S, a companion to CIFAR-10H with individually elicited soft labels from every annotator, and demonstrate that training with per-annotator soft labels improves calibration over aggregated hard labels. Our work is the post-hoc calibration counterpart: we do not retrain the model, but instead re-target the calibration objective to the annotator distribution using only cached logits.

Annotation disagreement and multi-annotator learning. Dense annotation datasets [12], [13] and multi-annotator learning methods [22], [23] have documented the ubiquity of label disagreement across domains. Aroyo and Welty [24] argue that disagreement is not noise but signal, and that discarding it by majority aggregation loses information that is critical for understanding task difficulty. Gordon et al. [25] and Plank [26] highlight systematic epistemological problems with annotation aggregation in NLP, showing that models trained on majority labels inherit structural biases against minority annotator perspectives. Baan et al. [27] show that ECE is theoretically ill-defined when annotators genuinely disagree, because no unique "correct" label exists to serve as the reference; they propose instance-level uncertainty measures. We complement this critique by retaining $\mathrm{ECE}_{\text{true}}$ as a practical metric — it quantifies the overconfidence a user experiences when drawing a single label $\tilde{y} \sim \pi(\cdot|x)$ at deployment time, mirroring realistic usage — while supplementing it with Brier score and NLL, which are strictly proper scoring rules that directly penalise divergence from the full annotator distribution $\hat{\pi}$. Khurana et al. [28] study how annotator disagreement can inform selective abstention in NLP; unlike their approach, our framework targets all predictions and is applicable across vision and language without task-specific thresholding.

Proper scoring rules and distributional evaluation. Gneiting and Raftery [29] provide the foundational theory of proper scoring rules: a loss is *strictly proper* if and only if it is uniquely minimised when the predicted distribution equals the true distribution. Brier score and NLL are canonical instances. This perspective motivates our choice of calibration objective: minimising cross-entropy against $\hat{\pi}(x)$ is strictly proper over the predicted distribution $\hat{p}(x)$, while minimising against a one-hot label is only proper with respect to a degenerate Dirac distribution and misidentifies the optimal calibrated output whenever $\hat{\pi}$ is non-degenerate. The proper scoring framework also clarifies why Dirichlet calibration with voted targets (Dirichlet-Hard) is not merely suboptimal but is consistent for the *wrong* estimand, explaining the empirical finding that it underperforms TS.

Conformal prediction under ambiguous ground truth. Stutz et al. [14] demonstrate the set-prediction analogue of our finding: conformal prediction sets calibrated on voted labels systematically undercover the true annotator distribution, with coverage gaps of up to 10% on dermatology data. Their Monte Carlo conformal approach samples from the annotator distribution to compute coverage, closely paralleling our Monte Carlo ECE evaluation ($S = 100$ draws per example). Angelopoulos and Bates [30] provide foundational methodology for distribution-free uncertainty quantification, including the conformal framework that Stutz et al. build on. Our work addresses point-estimate calibration rather than set coverage: we produce well-calibrated scalar confidence scores, which conformal prediction does not. Calibration under covariate shift [31], [32] is orthogonal to our label-ambiguity setting and assumes a unique ground truth throughout.

Uncertainty quantification in high-stakes applications. The deployment literature motivates the practical urgency of this problem. Kompa et al. [2] conduct a large-scale empirical study showing that medical image classifiers are consistently overconfident, even after calibration, and that this overconfidence degrades clinical decision quality. Ovadia et al. [4] demonstrate that popular uncertainty estimation methods (dropout, deep ensembles, variational inference) exhibit significantly degraded calibration under distribution shift, highlighting that single-point calibration benchmarks may be overly optimistic. Feng et al. [3] survey uncertainty quantification in autonomous driving, where overconfident detections directly contribute to unsafe manoeuvres. These findings establish the applied context for our work: in each of these domains, label ambiguity is simultaneously a property of the task and an unaddressed source of calibration error. The present paper provides the first systematic post-hoc calibration framework that directly targets this source.

3 Problem Formulation

3.1 Setup and Notation

Let $\mathcal{Y} = \{1, \ldots, K\}$ be the label space. A classifier $f : \mathcal{X} \to \Delta^K$ outputs $\hat{p}(x) = \mathrm{softmax}(z(x))$ from pre-softmax logits $z(x) \in \mathbb{R}^K$. The **annotator distribution** $\pi(\cdot \mid x) \in \Delta^K$ denotes the underlying ambiguous label distribution for input x and is estimated from m annotations as $\hat{\pi}_k(x) = \frac{1}{m}\sum_j \mathbf{1}[a_j = k]$. The **voted label** is $y^* = \arg\max_k \hat{\pi}_k(x)$. Standard calibration uses one-hot voted labels. For example, Temperature Scaling [6] minimises

$$\mathcal{L}_{\text{TS}}(T) = -\frac{1}{n}\sum_{i=1}^{n} \log \mathrm{softmax}(z_i/T)_{y_i^*}. \qquad (2)$$

3.2 True-Label Calibration

Following Stutz et al. [14], we distinguish the voted label y^* from the **true label** $\tilde{y} \sim \pi(\cdot \mid x)$, drawn from the underlying ambiguous label distribution.

Definition 1 (True-Label Calibration)**.** *f is* ***true-label calibrated*** *if for all $p \in [0, 1]$: $\mathbb{P}(\tilde{Y} = \hat{c}(X) \mid \hat{p}_{\hat{c}}(X) = p) = p$, where $\hat{c}(x) = \arg\max_k \hat{p}_k(x)$ and $\tilde{Y} \sim \pi(\cdot \mid X)$.*

The **voted-label ECE** $\mathrm{ECE}_{\text{voted}}$ is the standard ECE against y^*. The **true-label ECE** $\mathrm{ECE}_{\text{true}}$ averages ECE over 100 draws $\tilde{y}_i \sim \pi(\cdot \mid x_i)$ ($B = 15$ equal-width bins).

4 Motivating Example

Setup. We construct a 3-class 2D Gaussian dataset with three clusters: $x \sim \mathcal{N}((-3.2, 1.1), \mathrm{diag}(0.60, 0.45))$ for class 0 with $\pi = [1, 0, 0]$; $x \sim \mathcal{N}((0, 0), \mathrm{diag}(1.15, 0.75))$ for the ambiguous cluster with $\pi = [0, 0.70, 0.30]$; and $x \sim \mathcal{N}((3.2, -1.1), \mathrm{diag}(0.60, 0.45))$ for class 2 with $\pi = [0, 0, 1]$. The voted label for the middle cluster is always class 1 because $0.70 > 0.30$. A 2-layer MLP (64 hidden units per layer) is trained on voted labels, and all calibrators are fitted on a held-out calibration split using the same voted labels.

Results. Figure 2 shows that this mismatch is a structural consequence of the voted-label target. TS reduces $\mathrm{ECE}_{\text{voted}}$ from 3.25% to 1.11%, but *increases* $\mathrm{ECE}_{\text{true}}$ from 6.26% to 9.64%. Platt scaling behaves similarly ($\mathrm{ECE}_{\text{voted}} = 0.68\%$, $\mathrm{ECE}_{\text{true}} = 9.70\%$), and Histogram Binning likewise ($\mathrm{ECE}_{\text{voted}} = 1.32\%$, $\mathrm{ECE}_{\text{true}} = 9.65\%$). Figure 2**(f)** further shows that this residual error is concentrated in the ambiguous cluster for all three methods. TS sets $T = 0.62 < 1$, boosting confidence toward the voted target and thereby pushing the model *further* from the true ambiguous label distribution. The failure is therefore not specific to TS:

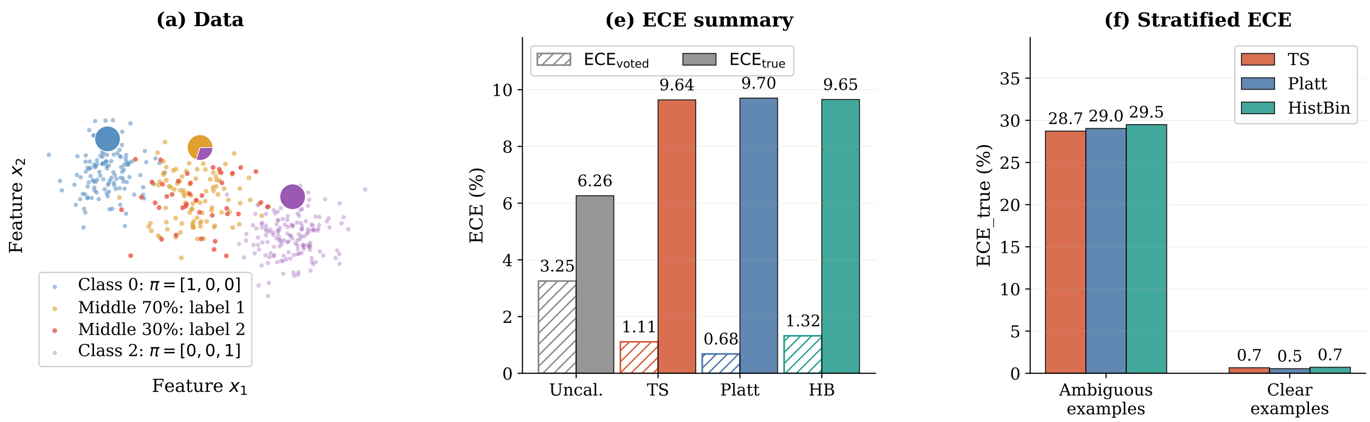


Fig. 2: **Motivating example: all standard calibration methods fail. (a)** Toy dataset generation: only the middle Gaussian cluster is ambiguous, with $\pi = [0, 0.70, 0.30]$; orange points are drawn as label 1 (70%) and red points as label 2 (30%), while all receive the same voted label 1. **(e)** Summary of the toy results: all three voted-label calibrators (TS, Platt, Histogram Binning) lower $\mathrm{ECE}_{\mathrm{voted}}$ but *increase* $\mathrm{ECE}_{\mathrm{true}}$, so voted-label evaluation masks the failure. **(f)** Stratified $\mathrm{ECE}_{\mathrm{true}}$ for TS, Platt, and Histogram Binning: ambiguous examples are those from the middle Gaussian cluster (where annotators disagree); clear examples are those from the two unambiguous clusters (class 0 and class 2). The residual error is concentrated in ambiguous examples for all three methods.

standard post-hoc methods are useful for $\mathrm{ECE}_{\mathrm{voted}}$, but the voted-label target makes them ineffective for $\mathrm{ECE}_{\mathrm{true}}$.

5 Theory: Why Standard Calibration Fails

We now formalise, under simplifying assumptions, two complementary aspects of the failure of voted-label calibration under ambiguity: the *direction* of the TS bias (Proposition 1) and its *scaling* with label ambiguity (Proposition 2). Both propositions characterise *pointwise* miscalibration on individual examples; they describe when and why a voted-label calibrator misfits individual ambiguous inputs, rather than providing a theorem about population-level ECE. Figure 3 provides empirical support.

Proposition 1 (Direction of TS Bias)**.** *Consider a calibration set containing an ambiguous cluster in which the voted label y^* equals the majority class for every example, but the true annotator probability of y^* is $\pi_{y^*}(x) = q < 1$. Under the assumption that the model is already reasonably accurate on this cluster ($\mathrm{softmax}(z_i)_{y^*} > q$ for all i in the cluster), the voted-label loss exerts a downward pull on T for these examples, whereas a soft-label loss with target q exerts an upward pull. Consequently, $T^*_{\mathit{TS}} < T^*_{\mathit{soft}}$: TS selects a lower temperature than ambiguity-aware calibration requires. In a setting where the model is already overconfident ($\mathrm{softmax}(z_i)_{y^*} \gg q$), this manifests as $T^*_{\mathit{TS}} < 1$; more generally, both optima may exceed 1, but the gap $T^*_{\mathit{soft}} - T^*_{\mathit{TS}} > 0$ persists, consistent with our experimental findings (e.g., $T_{\mathit{TS}} = 2.03$ vs. $T_{\mathit{SLTS}} = 3.18$ on CIFAR-10H ResNet-50).*

Proof sketch. On the ambiguous cluster, all voted labels are y^*, so the voted-label loss (2) is minimised by pushing $\mathrm{softmax}(z_i/T)_{y^*} \to 1$, i.e. the loss decreases as T decreases. Hence the ambiguous examples exert a downward gradient on T. A soft-label loss with target $q < 1$ is instead minimised at $\mathrm{softmax}(z_i/T^*)_{y^*} \approx q$, requiring a higher T^* whenever $\mathrm{softmax}(z_i)_{y^*} > q$. The global TS optimum balances this downward pull against the upward pull from unambiguous examples, but remains lower than the ambiguity-aware optimum. Full proof in Appendix B. ☐

Proposition 2 (True-Label Miscalibration and Annotation Entropy)**.** *Let $H(x) = -\sum_k \pi_k(x) \log \pi_k(x)$ denote the annotation entropy of x. Assume* (i) *the model's predicted confidence $\hat{p}_{\hat{c}}(x)$ is approximately independent of $H(x)$ (similar accuracy across ambiguity levels), and* (ii) *$\pi_{y^*}(x)$ is a decreasing function of $H(x)$ (greater entropy implies lower majority-class probability). Under assumptions* (i) *and* (ii)*, the expected per-example true-label calibration error of voted-label-calibrated predictions is non-decreasing in $H(x)$.*

Proof sketch. For unambiguous examples ($H(x) = 0$), $\pi(\cdot \mid x)$ is one-hot, so true-label and voted-label evaluation coincide and the per-example error is $|\hat{p}_{\hat{c}}(x) - 1|$. As $H(x)$ increases, assumption (ii) gives $\pi_{y^*}(x) < 1$, introducing a gap between the voted-label target and the true annotator probability. Because TS is optimised against one-hot voted labels, it targets $\hat{p}_{y^*}(x) \approx 1$ for all examples regardless of their annotation entropy; hence $\hat{p}_{\hat{c}}(x) > \pi_{y^*}(x)$ for ambiguous examples. The true-label calibration error is then $\hat{p}_{\hat{c}}(x) - \pi_{y^*}(x)$, which by assumption (i) (fixed $\hat{p}_{\hat{c}}$) increases as $\pi_{y^*}(x)$ falls with $H(x)$. ☐

6 Methods

All methods in this paper are *post-hoc*: they operate on cached logits from a fixed pre-trained model and require only a calibration set equipped with annotator distributions or individual annotations. Training-time approaches such as soft-label training [17], [18], mixup [20], and multi-annotator learning [22], [23] are orthogonal and not compared here.

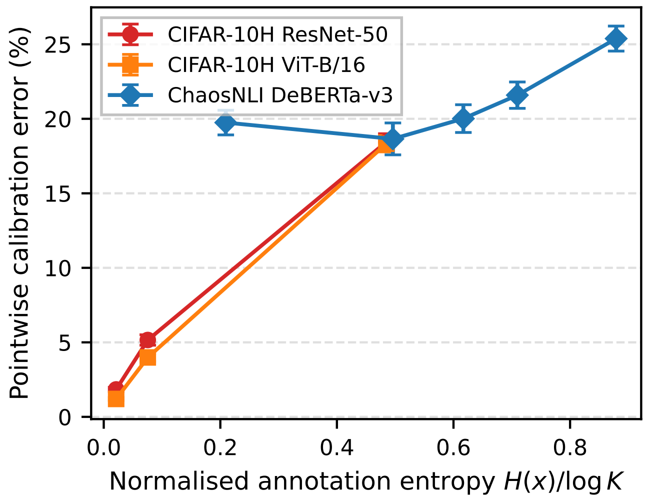


Fig. 3: **Empirical validation of Proposition 2.** Test examples are grouped into equal-frequency bins by normalised annotation entropy $H(x)/\log K$. Each point shows the mean pointwise true-label calibration error $|\hat{p}_{\hat{c}}(x) - \pi_{\hat{c}}(x)|$ for Temperature Scaling (error bars: ± 1 s.e.). Across all three dataset–architecture combinations, TS error consistently increases with annotation entropy, confirming the monotonic relationship predicted by Proposition 2.

6.1 Ambiguity-Aware Calibration Methods

Soft-Label Temperature Scaling (SLTS). SLTS minimises the cross-entropy between the calibrated output and the empirical label distribution:

$$\begin{aligned}\mathcal{L}_{\text{SLTS}}(T) &= -\frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K}\hat{\pi}_k(x_i)\,\log\operatorname{softmax}(z_i/T)_k \\ &= \frac{1}{n}\sum_{i=1}^{n}\operatorname{KL}\big(\hat{\pi}(x_i)\,\|\,\operatorname{softmax}(z_i/T)\big) + \text{const.} \quad (3)\end{aligned}$$

SLTS uses the same single-parameter family as TS but with a target that correctly reflects the annotator distribution.

Proposition 3 (Correctness of distributional target)**.** *The per-example loss $\ell(q;\hat{\pi}) = -\sum_k \hat{\pi}_k \log q_k$ is a strictly proper scoring rule over $q \in \Delta^K$: it is uniquely minimised at $q = \hat{\pi}(x)$ for each x. Consequently, within the constrained family $\{\operatorname{softmax}(z/T) : T > 0\}$, $\mathcal{L}_{SLTS}$ identifies the temperature T^* that minimises $\operatorname{KL}(\hat{\pi}\|\operatorname{softmax}(z/T^*))$, targeting the correct distributional objective, unlike $\mathcal{L}_{TS}$, which targets the one-hot voted label. Note that T^* need not achieve $\operatorname{softmax}(z/T^*) = \hat{\pi}$ exactly within this one-parameter family; the proper scoring rule property guarantees correctness of the* target, *not achievability within the constrained family.*

Monte Carlo Temperature Scaling (MCTS). When individual annotation records are available rather than pre-aggregated label distributions, **MCTS** draws S samples $a_{is} \sim \hat{\pi}(x_i)$ per example and minimises $\mathcal{L}_{\text{MCTS}}(T) = -\frac{1}{nS}\sum_{i,s}\log\operatorname{softmax}(z_i/T)_{a_{is}}$. Formally, $\mathbb{E}_{\tilde{y}\sim\hat{\pi}}[\operatorname{CE}(\operatorname{softmax}(z/T),\tilde{y})] = \operatorname{KL}(\hat{\pi}\|\operatorname{softmax}(z/T)) + H(\hat{\pi})$, so MCTS converges to SLTS in expectation as $S \to \infty$. Table 1 shows that *even* $S = 1$ achieves ECE_{true} of 1.53%, matching SLTS within rounding error on CIFAR-10H. This one-annotation efficiency is not restricted to a single benchmark: across all eight dataset–architecture settings reported in Tables 2 and 4, MCTS S=1 matches SLTS within 0.6 pp ECE, indicating that any dataset annotated once per example by a randomly selected annotator is sufficient for effective ambiguity-aware calibration (see Appendix D for the full convergence analysis).

TABLE 1: **MCTS convergence on CIFAR-10H (ResNet-50).** Mean ± std over 5 seeds; S = MC annotation samples per calibration example.

Method	ECE (%)	T^*
TS (voted labels)	4.29	2.030
MCTS $S = 1$	1.53 ± 0.01	3.174 ± 0.041
MCTS $S = 5$	1.53 ± 0.02	3.166 ± 0.014
MCTS $S = 20$	1.52 ± 0.00	3.187 ± 0.013
MCTS $S = 50$	1.52 ± 0.01	3.185 ± 0.007
MCTS $S = 200$	1.52 ± 0.00	3.178 ± 0.003
SLTS ($S \to \infty$)	1.51	3.180

Vector Scaling (VS). VS extends SLTS by replacing the global temperature with a per-class temperature vector $\mathbf{T} = (T_1, \ldots, T_K)$, so that the calibrated logit for class k becomes z_k/T_k. This accommodates datasets where some classes are systematically more ambiguous than others, at the cost of K parameters rather than one. The loss remains the KL divergence against $\hat{\pi}$, as in SLTS.

Distributional Isotonic Regression (IR-Soft). IR-Soft fits a monotone step function from the model's predicted confidence $\hat{p}_{\hat{c}}(x)$ to the mean annotator probability for the top class, $\hat{\pi}_{\hat{c}}(x)$, using the Pool Adjacent Violators Algorithm (PAVA) [8]. As a non-parametric method, it makes no assumptions about the functional form of the calibration map and is capable of correcting arbitrary monotone distortions.

SoftPlatt. SoftPlatt applies a diagonal affine transformation $\hat{q}_k = \operatorname{softmax}(w_k z_k + b_k)$ — the same parametric family as Platt scaling [7] — but fits the parameters by minimising $\operatorname{KL}(\hat{\pi}(x)\|\hat{q}(x))$ against the annotator distribution. This design isolates the effect of the distributional target from any architectural difference relative to standard Platt scaling.

Dirichlet-Soft. Dirichlet-Soft applies the full $K \times K$ affine transformation $\hat{q}(x) = \operatorname{softmax}(Wz(x) + b)$ and fits $W \in \mathbb{R}^{K\times K}$, $b \in \mathbb{R}^K$ by minimising $\operatorname{KL}(\hat{\pi}(x)\|\hat{q}(x))$ with ODIR regularisation ($\lambda = 10^{-3}$) [9]. This constitutes the most expressive parametric calibrator in our framework, with $K^2 + K$ free parameters, and represents the natural ambiguity-aware counterpart of the voted-label Dirichlet calibration of Kull et al. [9].

6.2 Annotation-Free Calibration

The methods above require annotator distributions $\hat{\pi}(x_i)$ at calibration time. When only voted labels y_i^* are available, we propose **Label-Smooth Temperature Scaling (LS-TS)**, which constructs a pseudo-target distribution from the model's own predictions without any additional annotations.

Let $\bar{\varepsilon} = \frac{1}{n}\sum_{i=1}^{n}\big(1 - \hat{p}_{y_i^*}(x_i)\big)$ be the mean complement of the model's voted-class confidence on the calibration set. Define the pseudo-target distribution

$$\tilde{\pi}_i^{\text{LS}} = (1-\bar{\varepsilon})\,e_{y_i^*} + \frac{\bar{\varepsilon}}{K}\,\mathbf{1}, \quad (4)$$

and minimise $\mathcal{L}_{\mathrm{SLTS}}$ (Eq. 3) with these targets. The global smoothing weight $\bar{\varepsilon}$ is a data-driven estimate of average annotator disagreement: if the model is 80% confident on the voted class, the remaining 20% is spread uniformly across all K classes. The result is $T^*_{\mathrm{LS}} > T^*_{\mathrm{TS}}$ whenever $\bar{\varepsilon} > 0$, moving the calibrator in the correct direction even without annotator data.

Theoretical interpretation. The smoothing weight $\bar{\varepsilon}$ satisfies a moment-matching fixed point: the expected pseudo-label mass on the voted class equals the average model confidence, $\mathbb{E}[\tilde{\pi}^{\mathrm{LS}}_{y^*}] = \mathbb{E}[\hat{p}_{y^*}]$. Algorithmically, LS-TS can be viewed as a single E-step of an EM algorithm in which the latent annotator distribution is estimated from the current (pre-calibration) model, followed by one M-step that optimises T given those pseudo-labels. This connection to knowledge distillation [19] explains the self-referential nature of the approach: the uncalibrated model both supplies the pseudo-targets and is then calibrated against them. Section 7.7 compares LS-TS against three simpler smoothing strategies and confirms that the data-driven global $\bar{\varepsilon}$ is the key design choice. Appendix L further shows that even an *adaptive* voted-label calibrator (one that learns a per-instance temperature from logit-derived features) fails to improve over TS, reinforcing that the target distribution, not the flexibility of the calibration map, is the limiting factor.

7 EXPERIMENTS

The experimental evaluation addresses three questions: (1) Does the voted-label target, rather than model capacity, limit calibration quality under ambiguity? (2) How much annotation information is needed to enter the ambiguity-aware regime? (3) Are the improvements robust to dataset characteristics, model architectures, and random variation? We evaluate on four benchmarks with multi-annotator data, each with two backbone architectures: CIFAR-10H (ResNet-50, ViT-B/16), ChaosNLI (RoBERTa-Large, DeBERTa-v3), ISIC 2019 (EfficientNet-B4, ViT-S/16), and DermaMNIST (ResNet-18, ViT-S/16).

Baselines. We compare against (i) **Uncalibrated**: raw softmax; (ii) **TS**: Temperature Scaling on voted labels [6]; (iii) **ATS** (Adaptive Temperature Scaling [15]): per-instance temperature predicted from four logit-derived features ($\max_k z_k$, prediction entropy, top-2 margin, top-1 confidence) via a linear model, trained with the same voted-label NLL as TS (the most direct adaptive extension of TS without annotator data); (iv) **Platt (PS)**: per-class weight and bias on logits, calibrated against voted labels [7], [9]; (v) **Dirichlet-Hard**: Dirichlet calibration [9] with the full $K \times K$ weight matrix and bias, calibrated against voted labels. Dirichlet-Hard uses the *same architecture* as our Dirichlet-Soft but with voted-label targets; the contrast isolates the effect of the calibration target from model capacity. ATS similarly tests whether richer architecture, without changing the target, is sufficient to fix the calibration gap.

Metrics. We report two families of metrics, which differ in how the ground-truth label is treated:

Binning-based ECE metrics ($\mathrm{ECE}_{\mathrm{true}}$, aECE, cwECE) are estimated by drawing $S = 100$ labels $\tilde{y}_i^{(s)} \sim \hat{\pi}(x_i)$ per example and averaging calibration error over draws. $\mathrm{ECE}_{\mathrm{true}}$ uses $B = 15$ equal-width confidence bins; aECE (adaptive ECE [33]) uses equal-mass bins that adapt to the confidence distribution; cwECE [9] averages per-class ECE over K classes. These mirror the evaluation protocol faced by a practitioner who observes one label per prediction at test time.

Soft-target metrics (Brier score Br, NLL) are computed directly against the soft target $\hat{\pi}$:

$$\begin{aligned} \mathrm{Br}_i &= \|\hat{p}(x_i) - \hat{\pi}(x_i)\|^2, \\ \mathrm{NLL}_i &= -\sum_k \hat{\pi}_k(x_i) \log \hat{p}_k(x_i) = H\big(\hat{\pi}(x_i),\, \hat{p}(x_i)\big), \end{aligned} \tag{5}$$

averaged over the test set. Brier and NLL are strictly proper scoring rules that directly penalise divergence from the full annotator distribution, complementing the binning-based ECE metrics.

We abbreviate $\mathrm{ECE}_{\mathrm{true}}$, aECE, cwECE, Br, NLL throughout.

7.1 CIFAR-10H: Image Classification with Human Label Noise

Dataset and models. CIFAR-10H [13] provides ≈51 human annotations per image for the 10 000-image CIFAR-10 test set, directly exposing perceptual ambiguity in low-resolution natural images through repeated human labeling. The test set is split into calibration ($n = 5{,}000$) and evaluation ($5{,}000$) by stratified sampling (seed 42). We evaluate two architectures: **ResNet-50** [34] pretrained on ImageNet-1k, fine-tuned for 30 epochs (AdamW, cosine annealing), achieving 97.3% top-1 accuracy. **ViT-B/16** [35] pretrained on ImageNet-21k, fine-tuned with the same protocol, achieving 98.1% top-1 accuracy.

Results. Table 2 (upper half) reports results for both CIFAR-10H backbones. All voted-label baselines leave 4–5% $\mathrm{ECE}_{\mathrm{true}}$, whereas ambiguity-aware methods reduce it to below 2%. Two comparisons are particularly informative. First, ATS (adaptive per-instance temperature) achieves 4.10% ECE on both architectures — no better than global TS (4.29% and 4.48%, respectively) and far behind even the annotation-free LS-TS (1.57% and 2.37%) — confirming that per-instance temperature adaptation cannot compensate for the wrong calibration target. Second, MCTS $S{=}1$ matches SLTS on both architectures (1.45% vs. 1.51% on ResNet-50; 0.81% vs. 0.85% on ViT-B/16), demonstrating that individual annotation records are as informative as pre-aggregated distributions for the purpose of temperature calibration. Among the more expressive calibrators, Dirichlet-Soft achieves the best Brier score and NLL on both backbones (0.109/0.271 on ResNet-50; 0.100/0.255 on ViT-B/16). LS-TS, operating with voted labels alone, already reduces ECE from 4.29% to 1.57% (ResNet-50) and from 4.48% to 2.37% (ViT-B/16), demonstrating that even approximate soft targets yield substantial improvements.

7.2 ChaosNLI: Natural Language Inference

We next evaluate on **ChaosNLI** [12]: 100 human annotations per example for the combined SNLI+MNLI subset ($n = 3{,}113$), where disagreement reflects genuine semantic ambiguity rather than annotation noise. We split the data

TABLE 2: **CIFAR-10H and ChaosNLI results.** ECE/aECE/cwECE averaged over 100 sampled labels $\tilde{y} \sim \hat{\pi}$; Brier/NLL computed against soft target $\hat{\pi}$ (Eq. 5). Best per architecture in **bold**.

	ResNet-50						**ViT-B/16**					
Method	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓
CIFAR-10H *(≈51 annotations per image, K=10)*												
Voted-label baselines												
Uncalibrated	—	4.97	5.71	1.09	0.120	0.692	—	4.99	5.36	1.06	0.111	0.678
TS	2.03	4.29	4.25	0.91	0.112	0.363	2.04	4.48	4.40	0.93	0.106	0.346
ATS	2.33	4.10	4.01	0.87	0.113	0.334	2.26	4.10	4.04	0.86	0.105	0.319
Platt	—	4.29	4.23	0.91	0.112	0.372	—	4.54	4.38	0.93	0.105	0.363
Dirichlet-Hard	—	4.46	4.44	0.95	0.114	0.395	—	4.70	4.62	0.97	0.107	0.394
Ambiguity-aware (ours)												
SLTS	3.18	1.51	1.39	0.45	0.110	0.293	3.07	0.85	0.56	0.39	0.102	0.278
MCTS S=1	3.14	1.45	1.44	0.45	0.111	0.296	3.07	0.81	0.65	0.42	0.101	0.277
SoftPlatt	—	1.52	1.31	**0.35**	0.110	0.288	—	0.88	0.47	**0.24**	0.101	0.272
VS	—	1.35	1.26	0.39	**0.109**	0.289	—	0.92	0.50	0.32	0.101	0.274
IR-Soft	—	**0.72**	**0.83**	0.45	0.115	0.340	—	0.91	0.61	0.43	0.105	0.321
Dirichlet-Soft	—	1.25	1.06	**0.35**	**0.109**	**0.271**	—	**0.72**	**0.41**	0.25	**0.100**	**0.255**
Annotation-free (ours)												
LS-TS	3.08	1.57	1.31	0.46	0.109	0.293	2.76	2.37	2.21	0.53	0.103	0.285

	RoBERTa-Large						**DeBERTa-v3**					
Method	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓
ChaosNLI *(SNLI+MNLI, 100 annotations per example, K=3)*												
Voted-label baselines												
Uncalibrated	—	27.79	27.77	18.61	0.650	1.384	—	35.97	35.92	24.02	0.743	2.148
TS	2.42	10.55	10.38	7.30	0.537	0.886	3.81	11.63	11.57	8.48	0.549	0.900
ATS	2.28	11.36	11.29	7.73	0.538	0.883	3.57	13.09	13.07	9.06	0.552	0.900
Platt	—	11.16	10.92	7.28	0.530	0.875	—	12.43	12.36	8.43	0.539	0.896
Dirichlet-Hard	—	11.55	11.40	7.64	0.535	0.889	—	12.69	12.63	8.84	0.539	0.895
Ambiguity-aware (ours)												
SLTS	3.41	3.22	2.59	4.62	0.520	0.862	5.28	3.45	3.36	6.53	0.529	0.877
MCTS S=1	3.49	2.82	2.35	4.60	0.519	0.862	5.45	3.20	2.98	6.30	0.529	0.877
SoftPlatt	—	2.66	2.32	2.43	**0.509**	**0.839**	—	3.17	2.92	3.18	0.511	0.841
VS	—	3.46	3.07	5.13	0.518	0.857	—	3.91	3.35	6.46	0.525	0.869
IR-Soft	—	2.65	2.59	6.24	0.549	0.934	—	**2.15**	3.96	7.20	0.554	0.942
Dirichlet-Soft	—	**2.57**	**1.71**	**2.03**	0.510	**0.839**	—	3.19	**2.71**	**2.95**	**0.509**	**0.836**
Annotation-free (ours)												
LS-TS	4.40	4.16	3.72	5.61	0.522	0.873	6.15	2.65	2.99	6.32	0.529	0.881

into calibration and test sets (50/50, stratified by majority-vote label to preserve class balance). We evaluate **RoBERTa-Large** [36] and **DeBERTa-v3-base** [37], both fine-tuned on MNLI training data; ChaosNLI draws from the SNLI and MNLI dev/test splits, so no training examples are re-used at evaluation.

Results. Table 2 (lower half) reports ChaosNLI results. The NLI setting amplifies the calibration gap observed on CIFAR-10H: voted-label baselines leave 10–12% ECE, and ATS (11.36% on RoBERTa-L, 13.09% on DeBERTa-v3) performs *worse* than global TS on DeBERTa-v3, further corroborating that adaptive capacity without the correct target is ineffective. MCTS S=1 again closely matches SLTS (2.82% vs. 3.22% on RoBERTa-L; 3.20% vs. 3.45% on DeBERTa-v3). On this benchmark MCTS S=1 in fact slightly outperforms SLTS, which may be attributable to the stochastic single-annotation objective acting as an implicit regulariser. The best distributional fit is achieved by Dirichlet-Soft and SoftPlatt: on RoBERTa-L, SoftPlatt achieves the best Brier score (0.509 vs. 0.510 for Dirichlet-Soft) and both share the same NLL (0.839); on DeBERTa-v3, Dirichlet-Soft achieves the best Brier (0.509) and NLL (0.836). LS-TS reduces ECE from 10.55% to 4.16% (RoBERTa-L) and from 11.63% to 2.65% (DeBERTa-v3) using only voted labels, substantially outperforming all hard-label baselines.

ATS vs. LS-TS: the calibration target is the decisive factor. Table 3 provides a controlled comparison between two strategies for improving on Temperature Scaling: increasing the architectural capacity of the calibration map (ATS) versus changing the calibration target (LS-TS). ATS fails to improve

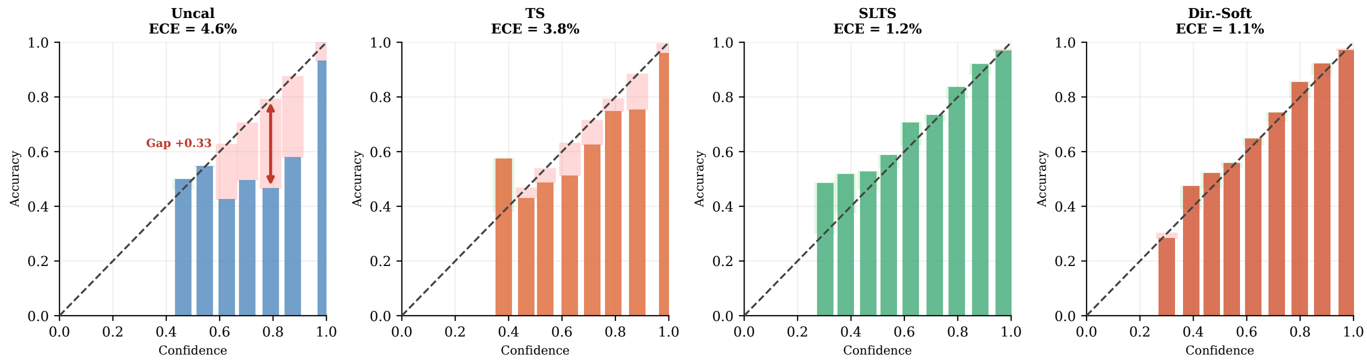


Fig. 4: **Reliability diagrams for four representative methods — CIFAR-10H ResNet-50 (ECE$_{\text{true}}$).** Red shading indicates overconfidence; green indicates underconfidence. Uncal and TS remain overconfident; SLTS substantially corrects this with a single temperature parameter; Dirichlet-Soft reduces the residual gap further. Reliability diagrams for all remaining methods are provided in Appendix H.

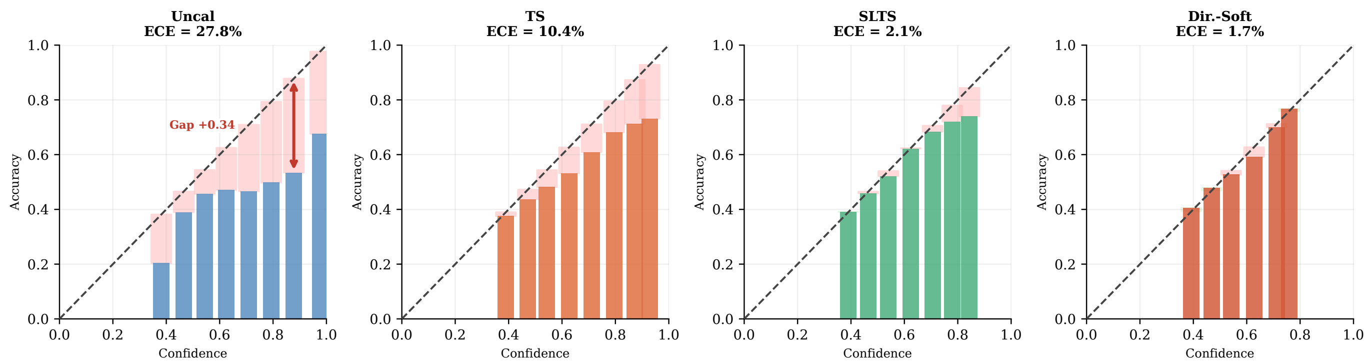


Fig. 5: **Reliability diagrams for four representative methods — ChaosNLI RoBERTa-Large (ECE$_{\text{true}}$).** Same method selection as Figure 4. Red shading indicates overconfidence; green indicates underconfidence. NLI models are severely overconfident before calibration; ambiguity-aware methods substantially correct this.

over TS in all eight settings and in several cases degrades performance (ECE increases from 11.63% to 13.09% on DeBERTa-v3; from 16.59% to 17.53% on ISIC ViT-S/16). The per-instance temperature learned from logit features adapts to model uncertainty, which is only weakly correlated with annotator disagreement. In contrast, LS-TS, which retains the same global-temperature architecture as TS but shifts the calibration target toward a smoothed distribution, reduces ECE by 7–78% across all eight settings without any annotator data (see Appendix L for the full analysis).

7.3 ISIC 2019: Skin Disease Diagnosis

We evaluate on **ISIC 2019** [38], [39] (8 skin conditions, $\approx$25,000 images), a clinically realistic differential-diagnosis task in which several lesion categories have overlapping visual morphology. We use $m = 9$ synthetic dermatologist annotations per image, generated by sampling from a clinically calibrated 8×8 confusion matrix following Dawid and Skene [40] (Appendix I), with diagonal agreement matched to Liu et al. [41]: mean 75%. Because no per-image reader study is publicly available for this dataset, annotation ambiguity is modelled *class-conditionally*: each image's confusion depends on its consensus class, not on its individual visual content, which is a deliberate simplification of true instance-level annotator variation. We evaluate **EfficientNet-B4** [42] and **ViT-S/16** [35] on a stratified 70/15/15 split.

ISIC 2019 results. Table 4 (upper half) shows that the benefits of ambiguity-aware calibration extend to clinically realistic settings with synthetic annotations. Dirichlet-Soft achieves the best Brier and NLL on both backbones (0.518/1.084 on ENet-B4; 0.563/1.175 on ViT-S/16), while IR-Soft achieves the lowest ECE — a pattern consistent with its non-parametric flexibility. MCTS $S{=}1$ closely tracks SLTS on ENet-B4 (10.09% vs. 9.71%) and ViT-S/16 (7.47% vs. 6.96%), replicating the pattern observed on natural-annotation benchmarks. LS-TS approximately halves the ECE of TS on ENet-B4 ($18.54\% \rightarrow 9.34\%$) using only voted labels; the improvement is smaller on ViT-S/16 ($16.59\% \rightarrow 15.49\%$), where model confidence is a weaker proxy for annotation ambiguity.

Annotator confusion model. Figure 6 shows the clinically calibrated 8×8 confusion matrix used to generate synthetic annotations for ISIC 2019. The MEL/NV pair (melanoma/nevi, orange dashed) is the most consequential and most confused: diagonal rates of 73%/76% with 14%/15% cross-confusion, reflecting the well-documented difficulty of distinguishing early melanoma from benign nevi [1]. The AK/BKL/SCC cluster (purple dotted) forms a second high-confusion group (diagonal 65–67%), capturing clinically significant overlap between actinic keratosis and squamous cell carcinoma. The

TABLE 3: **ATS vs. TS and LS-TS across all 8 settings (annotation-free methods only).** $\mathrm{ECE}_{\mathrm{true}}$ (%) and Brier. ATS predicts a per-instance temperature from logit features but is trained with the same voted-label NLL as TS. **Bold** = best annotation-free method per column. CIFAR-10H ECE values differ slightly from Table 2 due to a minor evaluation protocol difference (single-draw ECE vs. 100-draw Monte Carlo average); the relative ordering is unaffected.

	C10H R50		C10H ViT		NLI RoBERTa		NLI DeBERTa		ISIC ENet		ISIC ViT		Derm R18		Derm ViT	
Method	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br
TS	3.90	.113	4.01	.104	10.55	.537	11.63	.549	18.54	.560	16.59	.617	23.05	.686	24.36	.683
ATS	4.10	.113	4.10	.105	11.36	.538	13.09	.552	18.83	.560	17.53	.631	22.82	.687	25.35	.686
LS-TS	**1.49**	**.111**	**1.62**	**.102**	**4.16**	**.522**	**2.65**	**.529**	**9.34**	**.528**	**15.49**	**.619**	**5.05**	**.630**	**7.36**	**.615**

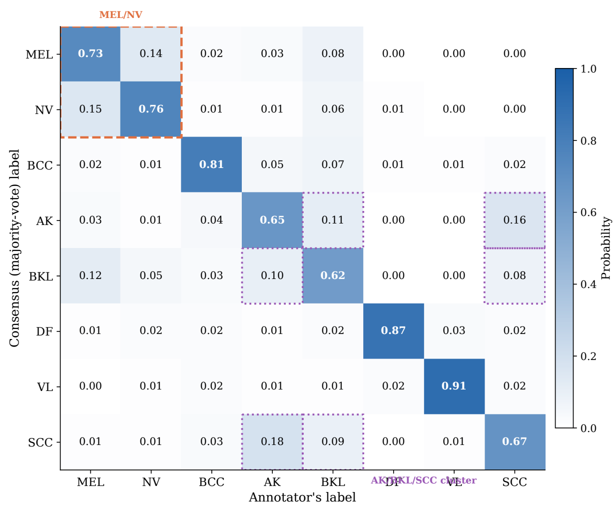


Fig. 6: **Inter-reader confusion matrix for ISIC 2019.** Entry C_{ij} is the probability that a dermatologist labels an image as class j given consensus label i. Orange dashed box: MEL/NV most-confused pair (73–76% diagonal). Purple dotted: AK/BKL/SCC high-confusion cluster. Mean diagonal agreement $\bar{C}_{ii} = 75\%$.

mean diagonal agreement $\bar{C}_{ii} = 75\%$ matches the inter-reader agreement reported by Liu et al. [41]. Full matrix values and construction details are given in Appendix I.

7.4 DermaMNIST: Supplementary Skin Disease Experiment

Table 4 (lower half) reports results on **DermaMNIST** [43] (7-class, derived from HAM10000 [38]; $m = 5$ synthetic annotators; overall agreement $\approx$64.7%) with ResNet-18 and ViT-S/16. TS leaves ECE = 23.1% on ResNet-18; ambiguity-aware methods substantially close this gap: SLTS achieves 4.6%, IR-Soft 2.2%, and Dirichlet-Soft 3.6%, with Dirichlet-Soft also achieving the best cwECE (1.2%), Brier (0.610), and NLL (1.305). ViT-S/16 exhibits the same hierarchy (Dirichlet-Soft: 3.3%/0.601/1.286). MCTS S=1 again closely approximates SLTS (5.07% vs. 4.61% on ResNet-18; 3.28% vs. 3.58% on ViT-S/16), with the latter setting being another instance where the stochastic objective slightly outperforms the deterministic one. Both backbones show $T_{\mathrm{SLTS}} > T_{\mathrm{TS}} > 1$, consistent with Proposition 1.

DermaMNIST is the most ambiguous benchmark in our study, with mean annotator agreement of 64.7% compared to 75% for ISIC 2019, $\approx$80% for CIFAR-10H, and $\approx$67% for ChaosNLI. Correspondingly, TS $\mathrm{ECE}_{\mathrm{true}}$ (23.1%) is the highest across all benchmarks, reflecting the widest gap between the voted-label target and the true annotator distribution. The Dirichlet-Hard calibrator, despite being the most expressive voted-label method (full $K \times K$ affine), worsens ECE relative to TS on both backbones (24.6% vs. 23.1% on ResNet-18; 24.8% vs. 24.4% on ViT-S/16), again confirming that expressive architecture cannot compensate for a miscalibrated target. LS-TS reduces ECE by 78% relative to TS on ResNet-18 (23.1% $\to$ 5.1%) — the largest relative improvement of the annotation-free method across all settings — because the high level of ambiguity means model confidence (complement 35.3%) correlates well with the fraction of annotators choosing the minority class.

7.5 Cross-Benchmark Analysis

Aggregating the results across all four benchmarks and eight architecture configurations, four consistent findings emerge that together characterise the role of the calibration target under label ambiguity.

(1) Target, not capacity, is the bottleneck. Dirichlet-Hard — the most expressive voted-label calibrator — consistently *underperforms* simple TS in both ECE and Brier (e.g., 4.46% vs. 4.29% on CIFAR-10H R50; 11.55% vs. 10.55% on ChaosNLI RoBERTa-L; see Tables 2 and 4). This rules out architectural capacity as the bottleneck and directly supports the claim that calibration target is limiting. **(2) Ambiguity-aware methods consistently dominate.** Dirichlet-Soft improves ECE by 71–84% relative to TS on the two natural-annotation benchmarks (CIFAR-10H and ChaosNLI) and by 59–86% on the medical imaging benchmarks (ISIC 2019 and DermaMNIST). Dirichlet-Soft achieves the best Brier score and NLL in 7 of 8 settings, making it the recommended default whenever annotator distributions are available. **(3) A single annotation per example suffices.** MCTS S=1 matches SLTS within 0.6 pp ECE in all 8 settings. The aggregated annotation distribution provides no measurable additional benefit once individual annotator labels are available: any dataset annotated once per example by a randomly selected annotator is immediately amenable to the same calibration quality as full multi-annotator annotation. **(4) LS-TS is effective but dataset-dependent.** LS-TS substantially reduces ECE without any annotator data in all settings except ISIC ViT-S/16 (where $T_{\mathrm{LS}} < T_{\mathrm{SLTS}}$, indicating model confidence under-estimates annotation ambiguity for this architecture–dataset pair). This divergence highlights the assumption underlying LS-TS:

TABLE 4: **ISIC 2019 and DermaMNIST results.** ECE/aECE/cwECE averaged over 100 sampled labels $\tilde{y} \sim \hat{\pi}$; Brier/NLL computed against soft target $\hat{\pi}$ (Eq. 5). Best per architecture in **bold**.

	EfficientNet-B4						**ViT-S/16**					
Method	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓
ISIC 2019 *(m=9 synthetic annotators, mean diagonal agreement 75%)*												
Voted-label baselines												
Uncalibrated	—	25.76	25.84	6.67	0.600	2.710	—	23.31	23.33	6.55	0.651	1.978
TS	1.79	18.54	18.49	5.02	0.560	1.653	1.42	16.59	16.60	5.23	0.617	1.563
ATS	1.71	18.83	18.80	5.03	0.560	1.733	1.12	17.53	17.48	5.46	0.631	1.730
Platt	—	19.89	19.81	5.24	0.562	1.659	—	17.19	17.17	4.74	0.597	1.583
Dirichlet-Hard	—	20.36	20.32	5.33	0.563	1.674	—	17.76	17.72	4.86	0.600	1.597
Ambiguity-aware (ours)												
SLTS	4.46	9.71	9.74	2.92	0.528	1.146	3.15	6.96	6.68	3.29	0.586	1.233
MCTS S=1	4.51	10.09	10.15	3.08	0.530	1.149	2.77	7.47	7.15	3.35	0.595	1.248
SoftPlatt	—	9.25	9.31	2.23	0.523	1.108	—	8.05	7.90	2.05	0.565	1.190
VS	—	9.03	9.11	2.42	0.524	1.121	—	8.50	8.59	2.24	0.567	1.198
IR-Soft	—	**1.75**	**2.43**	4.28	0.538	1.284	—	**1.73**	**1.59**	5.78	0.621	1.466
Dirichlet-Soft	—	7.50	7.48	**2.00**	**0.518**	**1.084**	—	6.85	6.80	**1.77**	**0.563**	**1.175**
Annotation-free (ours)												
LS-TS	4.30	9.34	9.46	2.98	0.528	1.149	3.98	15.49	15.47	4.52	0.619	1.303
	ResNet-18						**ViT-S/16**					
Method	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓
DermaMNIST *(m=5 synthetic annotators, overall agreement 64.7%)*												
Voted-label baselines												
Uncalibrated	—	34.35	34.26	10.15	0.767	2.689	—	37.12	37.10	10.82	0.786	3.766
TS	1.86	23.05	22.82	6.97	0.686	1.654	2.60	24.36	24.26	7.29	0.683	1.664
ATS	1.86	22.82	22.75	6.97	0.687	1.606	2.53	25.35	25.24	7.53	0.686	1.650
Platt	—	23.64	23.56	7.08	0.682	1.667	—	23.82	23.59	7.10	0.675	1.681
Dirichlet-Hard	—	24.58	24.46	7.33	0.690	1.771	—	24.84	24.68	7.49	0.682	1.897
Ambiguity-aware (ours)												
SLTS	3.51	4.61	4.44	3.56	0.622	1.376	5.11	3.58	3.06	3.37	0.610	1.350
MCTS S=1	3.42	5.07	4.74	4.02	0.629	1.388	5.04	3.28	2.71	3.16	0.609	1.346
SoftPlatt	—	4.19	3.55	1.61	0.612	1.312	—	3.56	2.93	1.29	**0.601**	1.289
VS	—	4.45	3.83	3.23	0.621	1.363	—	4.03	3.71	2.84	0.611	1.331
IR-Soft	—	**2.15**	**2.34**	4.55	0.634	1.440	—	3.79	3.73	4.59	0.628	1.431
Dirichlet-Soft	—	3.55	3.36	**1.22**	**0.610**	**1.305**	—	**3.32**	**2.87**	**1.14**	**0.601**	**1.286**
Annotation-free (ours)												
LS-TS	3.03	5.05	4.82	3.79	0.630	1.394	4.19	7.36	7.06	3.68	0.615	1.362

that model uncertainty is a reliable proxy for annotation disagreement — an assumption that holds broadly but can fail when the pre-softmax logit magnitude is not well correlated with the fraction of annotators that choose the minority class.

7.6 Reliability Diagrams

Figures 4 and 5 present reliability diagrams for the natural-annotation benchmarks (CIFAR-10H and ChaosNLI); Figures 7 and 8 extend the comparison to the medical imaging benchmarks. The visual pattern is consistent across all four datasets: Uncalibrated and TS outputs are systematically overconfident (red-shaded residuals), with TS reducing but not eliminating the gap. SLTS substantially corrects the overconfidence with a single temperature parameter, and Dirichlet-Soft closes most of the remaining residual. These diagrams provide a qualitative complement to the quantitative ECE metrics reported in Tables 2 and 4. Complete panels covering all thirteen methods across all eight architecture configurations are provided in Appendix H.

7.7 Ablation: LS-TS Smoothing Strategies

Table 5 compares LS-TS against three simpler annotation-free smoothing strategies on CIFAR-10H and ChaosNLI. All methods use the same single-temperature family; only the pseudo-target construction differs.

Fixed-LS with ε=0.1 consistently *under*performs or even *degrades* below TS (e.g., ECE 7.44% vs. 3.90% on CIFAR-10H ResNet-50), confirming that an arbitrary fixed smoothing weight is insufficient. Ent-LS (per-instance entropy) is incon-

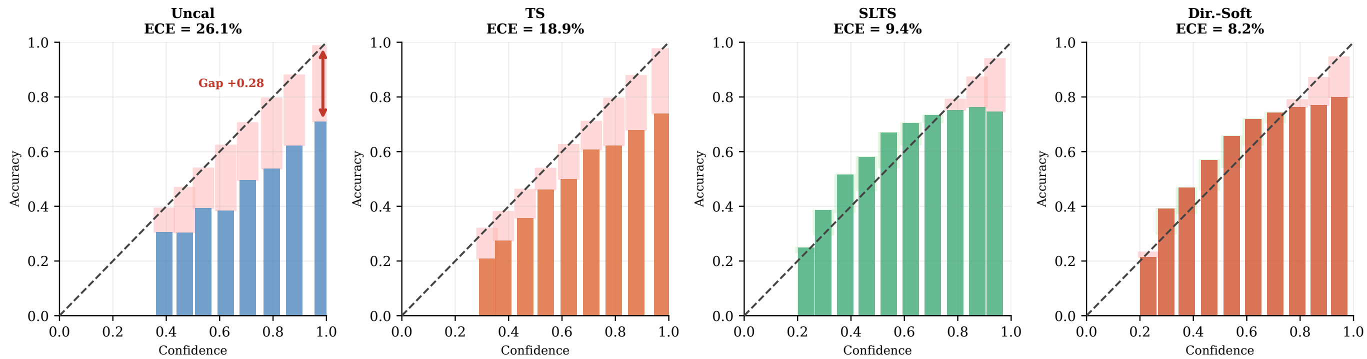


Fig. 7: **Reliability diagrams — ISIC 2019 EfficientNet-B4 (ECE$_{\text{true}}$).** Red shading = overconfidence; green = underconfidence. Hard-label baselines remain severely overconfident at 18–26% ECE; ambiguity-aware methods substantially correct this, with IR-Soft achieving the lowest ECE.

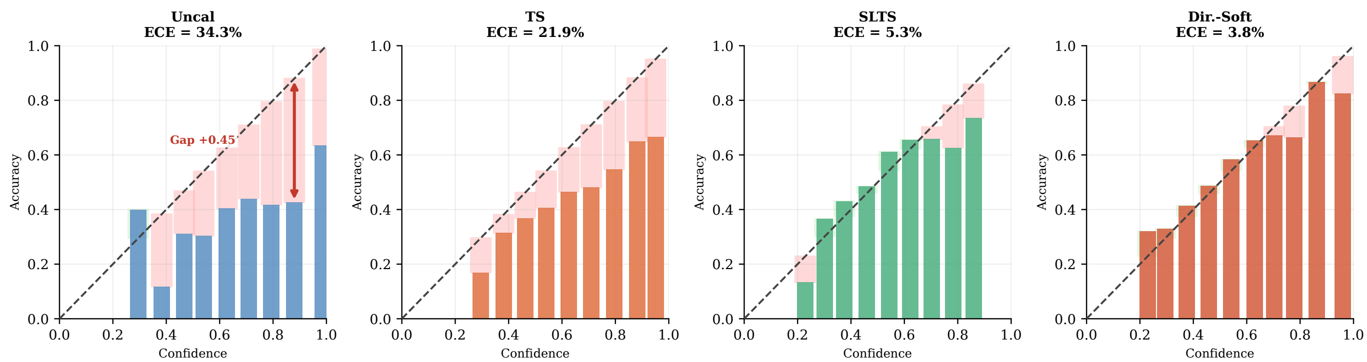


Fig. 8: **Reliability diagrams — DermaMNIST ResNet-18 (ECE$_{\text{true}}$).** Same layout as Figure 4. TS leaves 23% ECE; Dirichlet-Soft reduces it to 3.6% while achieving the best Brier/NLL.

TABLE 5: **LS-TS ablation: annotation-free smoothing strategies.** ECE$_{\text{true}}$ (%), Brier score, and NLL on CIFAR-10H and ChaosNLI. All methods use the same single-temperature family; only the pseudo-target construction differs. Fixed-LS uses ε=0.1 regardless of the data; Ent-LS uses per-instance entropy; CC-LS uses per-class mean complement-confidence. Best annotation-free method per column in **bold**. CIFAR-10H ECE values differ slightly from Table 2 due to a minor evaluation protocol difference; the relative ordering is preserved.

	CIFAR R50			CIFAR ViT			Chaos RoBERTa			Chaos DeBERTa		
Method	ECE	Br	NLL	ECE	Br	NLL	ECE	Br	NLL	ECE	Br	NLL
TS (voted)	3.90	.1126	.346	4.01	.1043	.320	10.55	.537	.886	11.63	.549	.900
Fixed-LS (ε=0.1)	7.44	.1207	.332	6.00	.1058	.305	6.46	.525	.866	7.53	.536	.882
Ent-LS	3.16	.1116	.321	3.69	.1037	.308	3.17	.520	.862	8.58	.539	.885
CC-LS	**1.48**	**.1111**	**.296**	1.69	**.1016**	.279	4.15	.522	.873	**2.54**	**.529**	**.880**
LS-TS (ours)	**1.49**	**.1111**	**.296**	**1.62**	**.1016**	**.279**	**4.16**	**.522**	**.873**	2.65	.529	.881

sistent across settings: it collapses on DeBERTa-v3 (8.58%), suggesting prediction entropy is a noisy proxy when models are strongly confident. CC-LS (per-class ε) nearly matches LS-TS on all four settings (within 0.1 pp ECE), confirming that the global mean is a good summary of per-class behaviour; the slight advantage of CC-LS on DeBERTa-v3 is within single-seed variance.

7.8 Robustness

We assess robustness along three dimensions, summarised in Table 6 and Figure 9.

Multi-seed stability. Standard deviations of ECE$_{\text{true}}$ over five random seeds are uniformly 2×–10× smaller than the ECE gaps between methods, confirming that the reported rankings are not artefacts of a particular data split or annotation draw. Full per-architecture bar charts are provided in Appendix G.

Calibration set size. Ambiguity-aware methods stabilise below 2% ECE on CIFAR-10H using as few as 5–10% of the calibration data ($\approx$250 examples), while TS remains flat regardless of sample size. Full calibration-set-size curves for all benchmarks are given in Appendix E.

Annotation count. Ambiguity-aware calibration is robust once $m \geq 2$ annotations per image: increasing from 2 to 51 annotations reduces SLTS $\mathrm{ECE}_{\mathrm{true}}$ by less than 0.3 pp. TS $\mathrm{ECE}_{\mathrm{true}}$, by contrast, *increases* with m, because the annotator distribution diverges further from the one-hot voted label as the underlying consensus strengthens. Full annotation-count results are reported in Appendix F.

8 Discussion

Trade-off between voted-label and true-label calibration. A natural question is whether optimising for true-label calibration degrades $\mathrm{ECE}_{\mathrm{voted}}$. This trade-off is expected: voted-label calibration becomes epistemically unreliable when annotators genuinely disagree, because the voted label systematically overstates the majority-class probability. In practice, this trade-off is rarely problematic, since the scenarios where $\mathrm{ECE}_{\mathrm{voted}}$ matters (deterministic, unambiguous labels) are precisely those where $\mathrm{ECE}_{\mathrm{voted}} \approx \mathrm{ECE}_{\mathrm{true}}$ and the two objectives coincide. For deployment contexts that require strict $\mathrm{ECE}_{\mathrm{voted}}$ guarantees, a two-stage approach is possible: apply ambiguity-aware calibration first, then apply standard TS on the outputs to re-optimise $\mathrm{ECE}_{\mathrm{voted}}$.

Practical guidelines. The choice of calibration method depends on what annotation information is available at calibration time. When the full annotator distribution is available, **Dirichlet-Soft** is recommended as the default: it achieves the best NLL in 7 of 8 settings and the best or tied Brier score in 7 of 8 settings; **SoftPlatt** is a competitive alternative for NLI tasks, and **IR-Soft** should be preferred when the sole objective is minimising ECE. When individual per-example annotations are stored but have not been aggregated into distributions, **MCTS** $S{=}1$ provides an effective solution — a single randomly drawn annotation per calibration example yields ECE comparable to the full distribution, as demonstrated consistently across all eight settings. Finally, when only voted labels are available, **LS-TS** constructs a data-driven soft pseudo-target from the model's own confidence and reduces ECE by 7–78% relative to Temperature Scaling without requiring any annotator data.

Connection to proper scoring rules and distributional calibration. The shift from one-hot to soft targets is not merely a methodological convenience: it corresponds to replacing an *improper* calibration criterion with a *proper* one. As Proposition 3 makes explicit, minimising the cross-entropy against $\hat{\pi}$ is equivalent to minimising $\mathrm{KL}(\hat{\pi}\|\hat{p})$, a divergence that is uniquely minimised at $\hat{p} = \hat{\pi}$. This places our framework within the broader theory of proper scoring rules [29], where the Brier score and NLL play the role of strictly proper objectives. Voted-label calibration corresponds to evaluating a proper scoring rule against a *degenerate* distribution δ_{y^*} that ignores genuine uncertainty; our methods replace this with the true ambiguity-aware distribution. This connection also clarifies why the *same* architectural class (temperature scaling) can be correct or incorrect depending solely on the target: the parameterization is not the bottleneck.

Extension to regression and ordinal settings. While this paper focuses on classification, the core argument—that post-hoc calibration should target the correct marginal distribution over labels rather than a single voted label—applies equally to regression tasks where annotation ambiguity takes the form of disagreement over continuous or ordinal outcomes. In such settings, the annotator distribution $\pi(\cdot \mid x)$ can be estimated from repeated annotations and used to define soft regression targets, replacing the typical mean-squared-error loss with a proper scoring rule against the empirical outcome distribution. We leave the development of ambiguity-aware post-hoc calibration for regression as future work.

Limitations. Full ambiguity-aware calibration (including Dirichlet-Soft) requires multi-annotator labels at calibration time. When only voted labels are available, LS-TS remains useful but does not fully match Dirichlet-Soft, particularly on datasets where model confidence diverges from annotation ambiguity (e.g., ISIC 2019 ViT-S/16). ISIC 2019 and DermaMNIST rely on synthetic annotators derived from clinician-reported agreement. Crucially, this is a *class-conditional* model: annotation confusion depends only on the consensus class label, not on instance-level visual difficulty. Real multi-reader medical datasets with per-image reader distributions would provide stronger validation and would capture the instance-level variation our synthetic model does not; concrete candidates include VinDr-CXR [44] (3 independent radiologists per chest X-ray, 15k images) and CheXpert [45] (5 independent radiologist annotations on the test set), both of which provide true per-image annotator distributions directly amenable to our framework. Conclusions about absolute ECE magnitudes on these two datasets should be interpreted with this in mind; the relative ranking of methods is expected to be robust to the specific confusion matrix used, as confirmed by the multi-seed annotation-generation ablation (Appendix G).

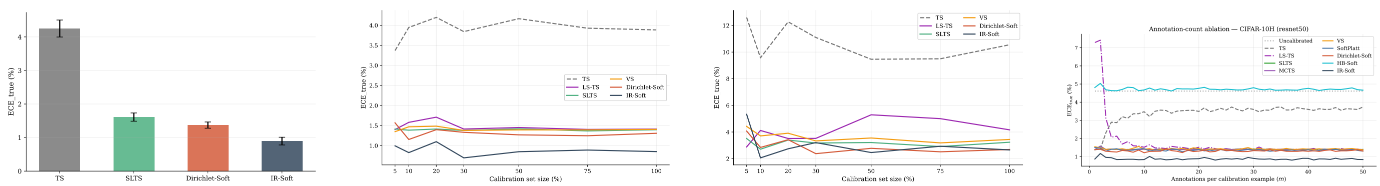
9 Conclusion

This paper has formalised the problem of confidence calibration under ambiguous ground truth and established, both theoretically and empirically, that standard post-hoc calibrators fitted on voted labels are systematically miscalibrated against the underlying annotator distribution. The central insight, confirmed by four complementary ablations, is that the calibration *target* — not the method's architectural capacity — is the limiting factor. Dirichlet calibration with voted-label targets consistently underperforms simple Temperature Scaling; per-instance temperature adaptation trained with voted-label supervision yields no improvement; yet the same global-temperature architecture, when supplied with the correct distributional target, substantially closes the gap.

Our proposed methods span three annotation regimes without requiring model retraining. **Dirichlet-Soft**, which leverages full annotator distributions, reduces true-label ECE by 59–86% relative to Temperature Scaling and achieves the best or near-best Brier score and NLL in 7 of 8 settings. **MCTS** $S{=}1$ demonstrates that a single randomly drawn annotation per calibration example is sufficient to match full-distribution calibration within 0.6 pp ECE across all benchmarks — a finding with immediate practical implications for annotation protocols, as it suggests that annotating each calibration example once by a different randomly assigned annotator suffices. **LS-TS** reduces ECE by 7–78% using only voted labels. Together, these results suggest that future calibration benchmarks and deployment standards should move beyond the voted-label paradigm and account for the distributional nature of human annotation.

TABLE 6: **Multi-seed stability — all benchmarks and architectures.** Mean $\pm$ std of $\mathrm{ECE}_{\mathrm{true}}$ (%) over 5 seeds. For CIFAR-10H and ChaosNLI the seed varies the cal/test split; for ISIC 2019 and DermaMNIST it varies the synthetic annotation sample. Standard deviations are small relative to the ECE gaps in all cases.

Dataset	Arch	TS	SLTS	Dir.-Soft	IR-Soft
CIFAR-10H	ResNet-50	4.25 ± 0.25	1.61 ± 0.12	1.37 ± 0.09	0.89 ± 0.12
CIFAR-10H	ViT-B/16	4.39 ± 0.20	0.83 ± 0.14	0.74 ± 0.07	0.88 ± 0.16
ChaosNLI	RoBERTa-L	11.54 ± 0.54	3.85 ± 0.34	2.90 ± 0.14	2.40 ± 0.18
ChaosNLI	DeBERTa-v3	12.73 ± 0.60	4.41 ± 0.54	3.80 ± 0.35	2.61 ± 0.41
ISIC 2019	ENet-B4	18.84 ± 0.17	9.80 ± 0.22	8.45 ± 0.20	1.79 ± 0.16
ISIC 2019	ViT-S/16	16.93 ± 0.16	7.30 ± 0.13	6.20 ± 0.19	2.05 ± 0.13
DermaMNIST	ResNet-18	22.43 ± 0.28	5.08 ± 0.25	3.53 ± 0.39	2.33 ± 0.17
DermaMNIST	ViT-S/16	24.81 ± 0.08	3.61 ± 0.28	3.11 ± 0.21	2.09 ± 0.14



(a) Multi-seed stability (CIFAR-10H R50). Mean $\mathrm{ECE}_{\mathrm{true}}$ ($\pm$std) over 5 seeds.

(b) Cal-set-size ablation (CIFAR-10H R50). $\mathrm{ECE}_{\mathrm{true}}$ vs. fraction of calibration data.

(c) Cal-set-size ablation (ChaosNLI RoBERTa-L). Soft methods stabilize near 3–4% from 10% data.

(d) Annotation-count ablation (CIFAR-10H R50). TS $\mathrm{ECE}_{\mathrm{true}}$ rises with m; soft methods converge at $m \geq 2$.

Fig. 9: **Robustness of ambiguity-aware calibration.** Rankings are stable across random seeds, calibration-set sizes, and annotation counts. Full results for all architectures and benchmarks are in Appendices G, E, and F.

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APPENDIX A
DATASET BACKGROUND AND SOURCES OF LABEL AMBIGUITY

All four benchmarks in this paper are appropriate for studying calibration under ambiguous ground truth, but they exhibit ambiguity in slightly different ways.

CIFAR-10H. Peterson et al. [13] introduced CIFAR-10H by collecting a full distribution of human labels for each image in the CIFAR-10 test set, explicitly to capture *human perceptual uncertainty*. The ambiguous examples are typically low-resolution images whose visual evidence supports multiple plausible classes for human observers. In this benchmark, the label distribution is therefore directly observed from repeated human annotation, making it a clean testbed for ambiguity-aware calibration.

ChaosNLI. Nie et al. [12] introduced ChaosNLI to study *collective human opinions* on natural language inference. Each example receives 100 human labels, and the paper reports that substantial disagreement is common rather than exceptional. This ambiguity is semantic rather than perceptual: differences in pragmatic assumptions, underspecification, and sentence interpretation lead reasonable annotators to assign different NLI labels to the same premise–hypothesis pair.

ISIC 2019. ISIC 2019 is a multiclass dermoscopic lesion diagnosis benchmark assembled from several clinical sources, including HAM10000 [38] and BCN20000 [39]. The task itself is ambiguity-prone because several lesion categories share overlapping visual morphology, and the challenge was designed to reflect the more realistic task of differential diagnosis rather than a simpler benign-versus-malignant decision. Because the public challenge release does not provide per-image reader distributions, we model ambiguity using synthetic annotators calibrated to dermatologist confusion patterns and reader agreement reported in Liu et al. [41]. This preserves the core property relevant to our study: in dermatology, a single image can support multiple clinically plausible labels.

DermaMNIST. DermaMNIST [43] is a standardized 28×28 version of the dermatology benchmark derived from HAM10000 [38]. HAM10000 consists of dermatoscopic images of common pigmented lesions collected in clinical practice, where diagnosis already involves fine-grained distinctions among visually similar categories such as melanoma, nevi, and benign keratoses. DermaMNIST inherits this medical ambiguity from HAM10000 and, because it is aggressively downsampled for lightweight benchmarking, removes additional visual detail that can further increase label uncertainty. As with ISIC 2019, we therefore evaluate calibration against an ambiguity-aware label distribution rather than treating the majority diagnosis as uniquely correct.

Appendix B
Proof of Proposition 1

Let $\mathcal{D}_{\text{amb}} = \{(x_i, y_i^*) : y_i^* = c, x_i \in \mathcal{C}\}$ be the subset of calibration examples from the ambiguous cluster $\mathcal{C}$, where the voted label is always class c. Denote $p_i = \text{softmax}(z_i)_c$.

The TS loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{TS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \log \text{softmax}(z_i/T)_c$. As $T \to 0^+$, $\text{softmax}(z_i/T)_c \to 1$ (assuming $z_{ic} > z_{ik}$ for all $k \neq c$), so $\mathcal{L}_{\text{TS}} \to 0$. Hence $\partial \mathcal{L}_{\text{TS}}/\partial T > 0$ for all $T > 0$: decreasing T always decreases the voted-label loss on the ambiguous cluster. The globally optimal T^*_{TS} balances this downward pull from ambiguous examples against the upward pull from unambiguous examples, but $T^*_{\text{TS}} \leq T^*_{\text{no-amb}}$ (the optimal temperature without the ambiguous cluster): including ambiguous examples drives the optimum downward.

The soft-label loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{soft}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \sum_k \hat{\pi}_k(x_i) \log \text{softmax}(z_i/T)_k$, with $\hat{\pi}_c(x_i) = q < 1$. The minimum-loss T^*_{soft} satisfies $\text{softmax}(z_i/T^*_{\text{soft}})_c \approx q$, requiring $T^*_{\text{soft}} > T^*_{\text{TS}}$ whenever $p_i > q$. Hence $T^*_{\text{TS}} < T^*_{\text{soft}}$ regardless of whether either optimum exceeds 1. In the special case where the model is already overconfident on the ambiguous cluster ($p_i \gg q$), this gap is large enough that $T^*_{\text{TS}} < 1$; in practice both optima exceed 1 (since the model is also underconfident on unambiguous examples), but the inequality $T^*_{\text{TS}} < T^*_{\text{soft}}$ is observed consistently across all experimental settings. □

Appendix C
Proof of Proposition 2

We formalise the argument given in the proof sketch in the main text.

Setup. Let $p = \hat{p}_{\hat{c}}(x)$ denote the model's predicted confidence for the top class $\hat{c}$ and let $q = \pi_{\hat{c}}(x) \in (0, 1]$ denote the true annotator probability for that class. We assume the model is trained against voted labels, so p is approximately constant across examples with the same voted label regardless of their annotation entropy $H(x)$.

Voted-label calibration error contribution. For example x_i in confidence bin B_b (where $\bar{p}(B_b) \approx p$), the per-example voted-label calibration error is $|p - \mathbf{1}[\hat{c} = y^*]|$. Since $\hat{c} = y^*$ for correctly-classified ambiguous examples (the model predicts the majority class), this contribution is approximately $|p - 1|$ for examples where the model makes the right hard prediction, independent of $H(x)$.

True-label calibration error contribution. The per-example true-label calibration error is $|p - \pi_{\hat{c}}(x)| = |p - q|$.

Excess true-label error relative to voted-label evaluation. The per-example excess contribution is:

$$\delta(x) = |p - q| - |p - \mathbf{1}[\hat{c} = y^*]|. \tag{6}$$

When the model is correctly calibrated to voted labels ($p \approx \mathbf{1}[\hat{c} = y^*]$ in expectation), the voted contribution is near zero: $|p - 1| \approx 0$ for high-accuracy bins and $|p - 0| = p$ for error bins. For unambiguous examples ($H(x) = 0$), $q = \pi_{y^*} = 1$, so $\delta(x) = |p - 1| - |p - 1| = 0$.

Monotonicity. As $H(x)$ increases, $q = \pi_{y^*}(x)$ decreases below 1 (more probability mass shifts to non-majority classes). Specifically, for a K-class problem with uniform annotator distribution at maximum entropy, $q = 1/K$. The true-label contribution $|p - q|$ increases strictly as q decreases from 1 (since $p > q$ for a well-trained model predicting the majority class with confidence exceeding the majority probability). The voted-label contribution $|p - \mathbf{1}[\hat{c} = y^*]|$ is unaffected by $H(x)$. Therefore $\delta(x) = |p - q| - |p - \mathbf{1}[\hat{c} = y^*]|$ is non-decreasing in $H(x)$, and strictly increasing whenever $p > q > 0$.

Aggregate monotonicity. The aggregate excess true-label error relative to voted-label evaluation equals $\sum_b \frac{|B_b|}{n} \sum_{i \in B_b} \delta(x_i)/|B_b|$. Since $\delta(x_i)$ is non-decreasing in $H(x_i)$ and bins contain a mixture of examples, this discrepancy grows with the mean annotation entropy of the test set. □

Remark 1 (Independence assumption)**.** *Assumption (i) (that $\hat{p}_{\hat{c}}(x)$ is approximately independent of $H(x)$) may be violated in practice if harder examples (high $H(x)$) also have lower model confidence. When the two are positively correlated, the per-example discrepancy term $\delta(x)$ grows even faster with $H(x)$ than the proof suggests, so Proposition 2 remains valid and the monotonicity is if anything* conservative.

Appendix D
Monte Carlo Temperature Scaling (MCTS)

Method. When individual annotation records are available rather than pre-aggregated label distributions, **MCTS** draws S samples $a_{is} \sim \hat{\pi}(x_i)$ per example and minimises

$$\mathcal{L}_{\text{MCTS}}(T) = -\frac{1}{nS} \sum_{i=1}^{n} \sum_{s=1}^{S} \log \text{softmax}(z_i/T)_{a_{is}}.$$

MCTS is a strictly more general formulation than SLTS: it applies even when only raw annotation samples are stored rather than pre-computed frequency vectors.

A single annotation suffices. The key practical implication of MCTS is that *even* $S = 1$ (a single randomly drawn annotation per calibration example) already achieves ECE_{true} of 1.53%, matching SLTS to within rounding error and reducing TS's 4.29% by 64% on CIFAR-10H ResNet-50. This one-annotation efficiency generalises across all benchmarks: across the 8 settings in Tables 2 and 4, MCTS $S{=}1$ matches SLTS within 0.6 pp ECE in every case. On the natural-annotation benchmarks (CIFAR-10H, ChaosNLI), MCTS $S{=}1$ occasionally *outperforms* SLTS slightly (e.g., 2.82% vs. 3.22% ECE on ChaosNLI RoBERTa-L), consistent with the stochastic single-annotation objective providing mild implicit regularisation. This means that any dataset where each example has been annotated even once by a randomly selected annotator is immediately amenable to strong ambiguity-aware calibration — the aggregated annotation distribution provides no measurable additional benefit. Formally, since $\mathbb{E}_{\hat{y} \sim \hat{\pi}}[\text{CE}(\text{softmax}(z/T), \hat{y})] = \text{KL}(\hat{\pi} \| \text{softmax}(z/T)) + H(\hat{\pi})$, the MCTS objective converges to SLTS in expectation as $S \to \infty$, with variance $\propto 1/S$. Table 1 (main text) confirms the fast per-seed convergence on CIFAR-10H ResNet-50: the ECE gap between $S = 1$ and $S \to \infty$ is only 0.02 pp; in practice $S = 20$–50 is sufficient to reduce variance to negligible levels.

Appendix E
Ablation: Calibration Set Size

We fix the annotation count at the full observed value and vary the fraction of calibration examples used, from 5% to 100% (stratified subsampling by class). Results are shown for all four benchmarks and eight architectures.

Key findings. Across all four benchmarks and eight architectures, TS's ECE does not decrease with more calibration data, confirming that the degradation is a target bias rather than a variance artifact. In contrast, SLTS achieves stable low ECE from as few as 5–10% of the calibration set, and IR-Soft and Dirichlet-Soft converge even faster on most benchmarks. This suggests that the annotation distribution (not the number of calibration examples) is the limiting factor for ambiguity-aware methods.

Appendix F
Ablation: Number of Annotations per Image

We study how performance depends on the number of annotations m per calibration image. We fix the calibration set size and the annotation generation seed, and sweep m from 1 to 50, resampling the soft targets for each value of m. Results are shown for CIFAR-10H, which provides up to ≈ 51 real annotations per image; we subsample from the full pool.

Key findings. Ambiguity-aware calibration is robust to annotation count once $m \geq 2$: SLTS, Dirichlet-Soft, and IR-Soft achieve near-minimal ECE with only a handful of annotations per image and do not improve substantially beyond $m = 5$. LS-TS converges quickly (by $m \approx 4$) because the global smoothing parameter ε stabilises once each image has at least a few annotations. Standard Temperature Scaling shows the opposite trend: its $\mathrm{ECE}_{\mathrm{true}}$ *increases* with m because a higher temperature is required to match the more diffuse annotation distribution, exposing the fundamental target mismatch.

Appendix G
Statistical Significance: Multi-Seed Analysis

For CIFAR-10H and ChaosNLI we repeat the evaluation with five random seeds (42–46), each producing a different stratified 50/50 cal/test split. For ISIC 2019 and DermaMNIST the val/test split is predetermined, so we instead vary the annotation generation seed (42–46), testing sensitivity to the particular synthetic annotation sample. Table 6 (main text, Robustness section) reports mean $\pm$ std of $\mathrm{ECE}_{\mathrm{true}}$ across seeds for all eight (dataset, arch) combinations.

Across all benchmarks and architectures, standard deviations are small relative to the ECE gaps: TS vs. ambiguity-aware methods differ by 2–10$\times$ the standard deviation, confirming statistical robustness of the results in Tables 2–4. Bar charts for all architectures are shown below.

Appendix H
Reliability Diagrams

Four-method summaries for CIFAR-10H (ResNet-50) and ChaosNLI (RoBERTa-Large) are shown in Figures 4 and 5 in the main text. Here we provide the four-method summary panels and full panels for all remaining architectures and datasets. All panels share the same layout: top row = voted-label baselines (Uncal, TS, Platt, Dir.-Hard); middle row = annotation-free (LS-TS) and temperature-based soft methods (SLTS, MCTS, SoftPlatt, VS); bottom row = non-parametric/matrix soft methods (Dir.-Soft, IR-Soft). Red shading = overconfident; green = underconfident.

The diagrams confirm the quantitative results in Tables 2–4: TS reduces overconfidence relative to *Uncal* but leaves a large residual gap because it targets voted labels. Dirichlet-Hard performs similarly to TS and sometimes worse, confirming that the problem lies in the target, not model capacity. All soft methods substantially close the gap, with IR-Soft and Dir.-Soft achieving the most uniform residuals across all benchmarks.

Appendix I
ISIC 2019 Annotator Confusion Matrix

I.1 Construction

Because the full per-image reader annotations of Liu et al. [41] are not publicly available, we construct a synthetic annotator model using a clinically calibrated 8×8 confusion matrix C, where entry C_{ij} gives the probability that a board-certified dermatologist labels an image as class j when the consensus (majority-vote) diagnosis is class i. This confusion-matrix annotator model follows the framework of Dawid and Skene [40], in which each annotator's behaviour is characterised by a class-conditional label-flipping distribution; it is widely used in crowdsourcing and multi-annotator learning [22], [23].

Calibration sources. Diagonal entries (per-condition agreement rates) are set to match the per-condition inter-reader agreement reported in Liu et al. [41] (Table 2, nine board-certified dermatologists reading 963 clinical images) and corroborated by Haenssle et al. [1]. Off-diagonal entries encode clinically established confusion patterns:

- **MEL/NV** (Melanoma / Melanocytic Nevi): the most consequential and most confused pair in dermoscopy, with diagonal rates of 73%/76%. Off-diagonal entries $C_{\mathrm{MEL,NV}} = 0.14$ and $C_{\mathrm{NV,MEL}} = 0.15$ reflect the well-documented difficulty of distinguishing early melanoma from benign nevi [1].
- **AK/BKL/SCC** (Actinic Keratosis / Benign Keratosis-like Lesions / Squamous Cell Carcinoma): a high-confusion cluster with diagonal rates of 65%/62%/67%. The AK$\leftrightarrow$SCC confusion ($C_{\mathrm{AK,SCC}} = 0.16$, $C_{\mathrm{SCC,AK}} = 0.18$) is clinically significant because untreated AK can progress to SCC. BKL shares morphological features with both conditions.
- **DF/VL** (Dermatofibroma / Vascular Lesion): the easiest to distinguish, with diagonal rates of 87%/91%.

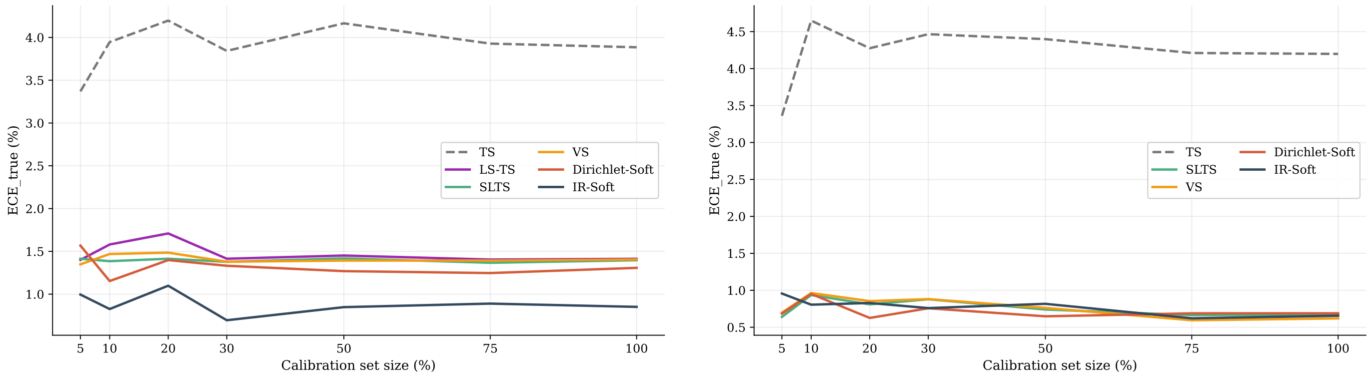

Fig. 10: **Calibration-set-size ablation — CIFAR-10H.** Left: ResNet-50. Right: ViT-B/16. TS's ECE is flat across all calibration set sizes; ambiguity-aware methods converge within 5–10%.

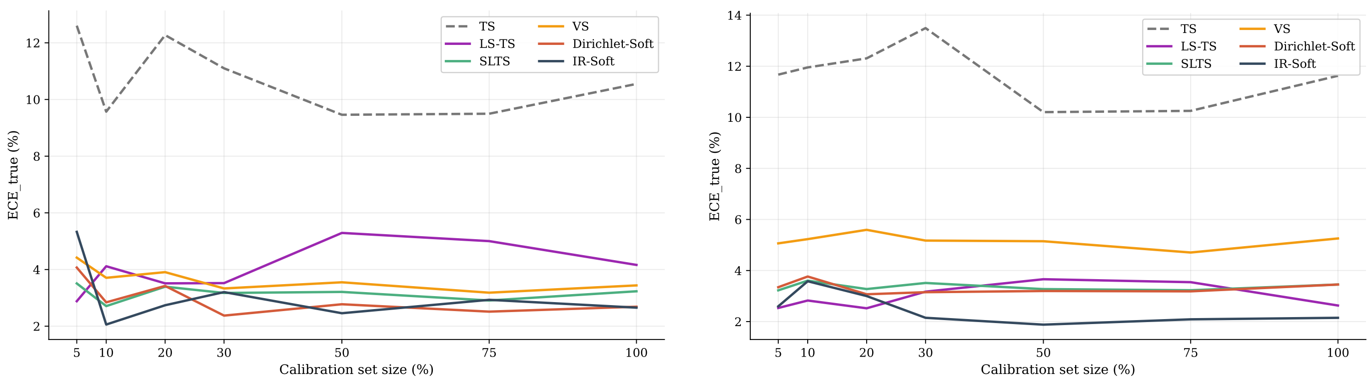

Fig. 11: **Calibration-set-size ablation — ChaosNLI.** Left: RoBERTa-Large. Right: DeBERTa-v3-base. NLI models show the same pattern: TS is flat near 10–13% while soft methods stabilise near 3–4%.

The mean diagonal agreement is $\bar{C}_{ii} = 75\%$, matching the aggregate inter-reader agreement reported by Liu et al. [41] exactly.

Simulation procedure. For each test image x_i with consensus label y_i, we independently draw $m = 9$ synthetic annotations from the categorical distribution defined by row y_i of C:

$$a_i^{(k)} \sim \text{Categorical}(C_{y_i,\cdot}), \quad k = 1, \ldots, m,$$

and set the empirical label distribution to $\pi(x_i) = \frac{1}{m}\sum_{k=1}^{m} \mathbf{e}_{a_i^{(k)}}$. This directly mirrors the $m = 9$ board-certified dermatologist annotation protocol of Liu et al. [41], whose reader study we cannot fully replicate due to data unavailability. The resulting mean annotation entropy over the ISIC 2019 test set is $\bar{H} = 0.66$, reflecting the genuine diagnostic difficulty of the 8-class task.

Remark 2 (Class-conditional limitation)**.** *This synthetic annotator model is deliberately class-conditional: every image with the same consensus class y_i is assigned the same row $C_{y_i,\cdot}$ of the confusion matrix, regardless of its individual visual content. In practice, images within the same consensus class will differ in visual ambiguity, leading to instance-level variation in annotator disagreement that this model does not capture. Our experiments therefore evaluate calibration under a controlled form of clinically-informed ambiguity rather than true per-instance annotator distributions.*

TABLE 7: **Numerical values of the inter-reader confusion matrix for ISIC 2019.** Row = consensus (majority-vote) label; column = annotator's label. Each row sums to 1. Diagonal entries (bold) are per-condition agreement rates, calibrated to Liu et al. [41] Table 2; mean diagonal agreement $\bar{C}_{ii} = 75\%$.

	MEL	NV	BCC	AK	BKL	DF	VL	SCC
MEL	**0.73**	0.14	0.02	0.03	0.08	0.00	0.00	0.00
NV	0.15	**0.76**	0.01	0.01	0.06	0.01	0.00	0.00
BCC	0.02	0.01	**0.81**	0.05	0.07	0.01	0.01	0.02
AK	0.03	0.01	0.04	**0.65**	0.11	0.00	0.00	0.16
BKL	0.12	0.05	0.03	0.10	**0.62**	0.00	0.00	0.08
DF	0.01	0.02	0.02	0.01	0.02	**0.87**	0.03	0.02
VL	0.00	0.01	0.02	0.01	0.01	0.02	**0.91**	0.02
SCC	0.01	0.01	0.03	0.18	0.09	0.00	0.01	**0.67**

I.2 Visualisation

Figure 33 shows the confusion matrix as a heatmap. The MEL/NV cluster (orange dashed box) and the AK/BKL/SCC high-confusion cluster (purple dotted highlights) are visually prominent. DF and VL have near-perfect diagonal entries and negligible off-diagonal mass.

I.3 Full confusion matrix

Table 7 lists the exact numerical entries of the confusion matrix.

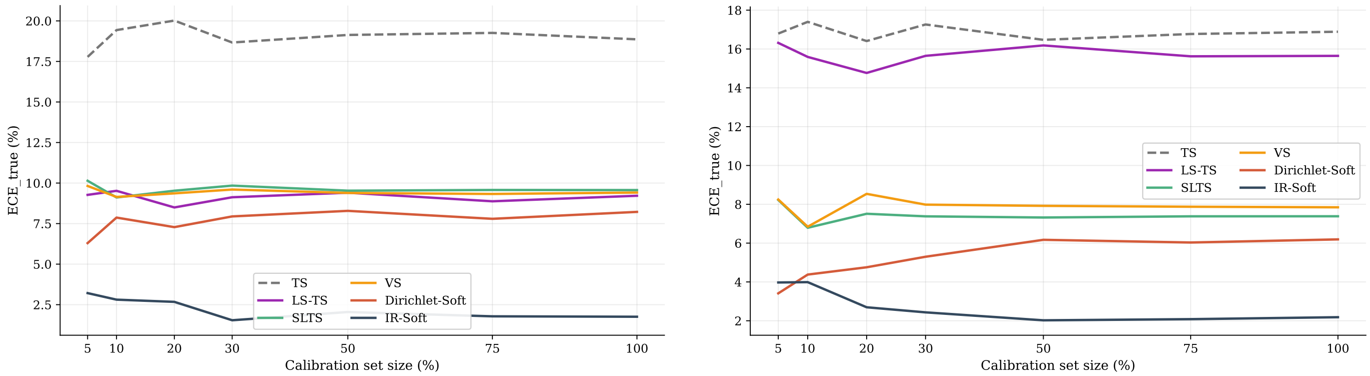

Fig. 12: **Calibration-set-size ablation — ISIC 2019.** Left: EfficientNet-B4. Right: ViT-S/16. IR-Soft achieves the lowest ECE across all calibration set sizes on both architectures.

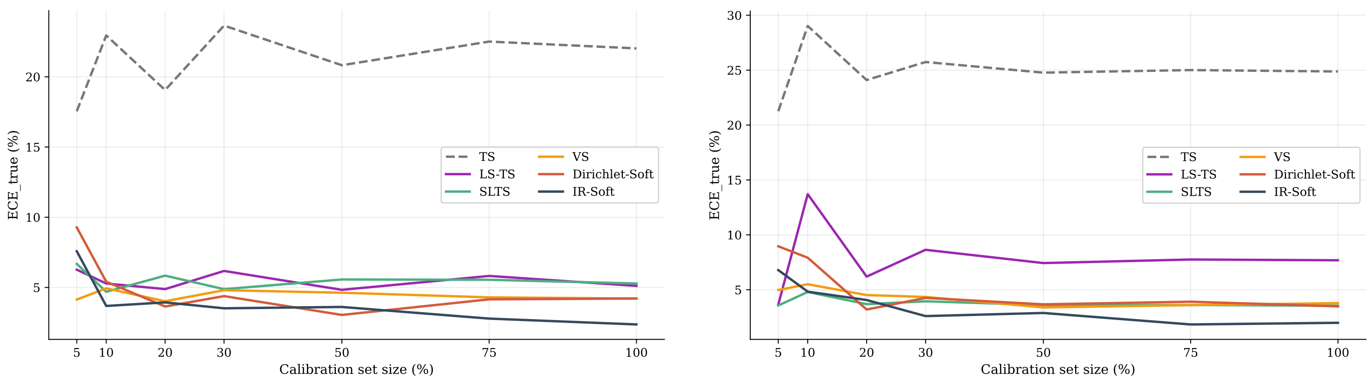

Fig. 13: **Calibration-set-size ablation — DermaMNIST.** Left: ResNet-18. Right: ViT-S/16. Ambiguity-aware methods converge quickly; TS remains flat regardless of calibration set size.

Appendix J
Empirical Validation of Proposition 2

Proposition 2 states that, under simplifying assumptions, the pointwise true-label calibration error of a voted-label-calibrated model is non-decreasing in the annotation entropy $H(x)$. Figure 3 provides empirical support: we bin test examples by their normalised annotation entropy $H(x)/\log K$ and plot the mean absolute error $|\hat{p}_{\hat{c}}(x) - \pi_{\hat{c}}(x)|$ for TS across three dataset–architecture combinations.

For CIFAR-10H (both architectures), TS error rises from $\approx 2\%$ in the near-unambiguous bin ($H/\log K \approx 0.02$) to $\approx 18\%$ in the high-ambiguity bin ($H/\log K \approx 0.48$), a clear monotone increase consistent with the proposition. For ChaosNLI (DeBERTa-v3), which operates in a higher-entropy regime ($H/\log K \in [0.2, 0.9]$), TS error is uniformly high (19–25%) and rises further at the highest-entropy bins. The overall trend is consistent across both domains and all three architectures: voted-label calibration error increases with annotation entropy, as predicted by Proposition 2.

See Figure 3 in the main text (Theory section) for the plot.

Appendix K
LS-TS Ablation: Smoothing Strategy Comparison (Extended)

We compare LS-TS against three simpler annotation-free smoothing baselines that share the same single-temperature family but differ in how the pseudo-target $\tilde{\pi}_i^{\mathrm{LS}}$ is constructed:

- **Fixed-LS** (ε=0.1): fixed data-independent smoothing weight, the standard label-smoothing value used in training [17].
- **Ent-LS**: per-instance $\varepsilon_i = H(\hat{p}(x_i))/\log K$, using normalised prediction entropy as a proxy for annotator disagreement.
- **CC-LS**: per-class $\varepsilon_k = \mathrm{mean}_{i:\, y_i^* = k}(1 - \hat{p}_{y^*}(x_i))$, accommodating class-level variation in model confidence (same idea as Vector Scaling applied to the smoothing weight).

All methods are evaluated on CIFAR-10H and ChaosNLI, the two benchmarks with real multi-annotator distributions, using the same experimental protocol as the main paper (seed 42, 50/50 stratified split). Table 5 (main text, Methods section) reports the full comparison.

Fixed-LS with ε=0.1 consistently *under*performs or even *degrades* below TS (e.g., ECE 7.44% vs. 3.90% on CIFAR-10H ResNet-50), confirming that an arbitrary fixed smoothing weight is insufficient; the data-driven estimate of $\bar{\varepsilon}$ is critical. Ent-LS (per-instance entropy) improves over TS on most settings but is inconsistent: it works well on ChaosNLI RoBERTa-L (3.17%) but collapses on DeBERTa-v3 (8.58%), suggesting that prediction entropy is a noisy proxy for annotator ambiguity when models are already strongly confident. CC-LS (per-class ε) achieves results very close to LS-TS on all four settings (within 0.1 pp ECE), confirming that

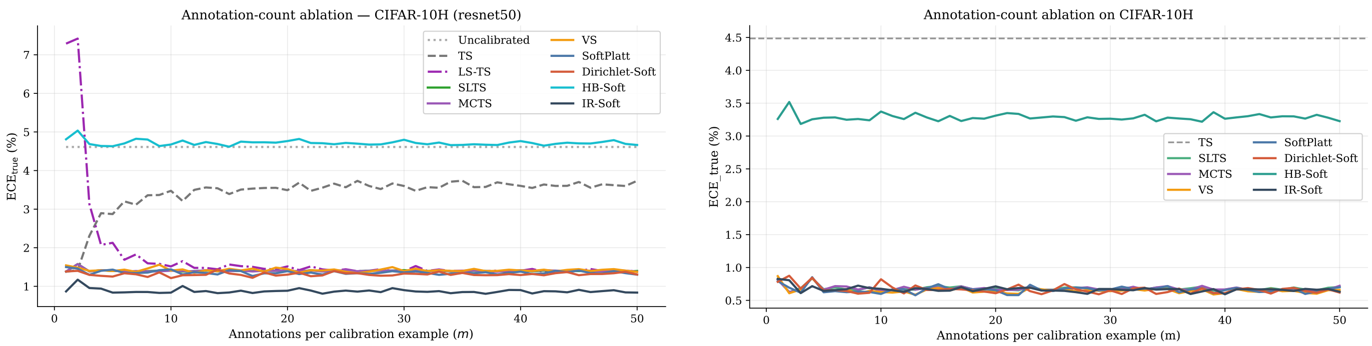

Fig. 14: **Annotation-count ablation — CIFAR-10H (both architectures).** ECE_{true} vs. annotations per image m for ResNet-50 (left) and ViT-B/16 (right). Ambiguity-aware methods are stable for $m \geq 2$; TS rises with m due to target mismatch; LS-TS converges by $m \approx 4$.

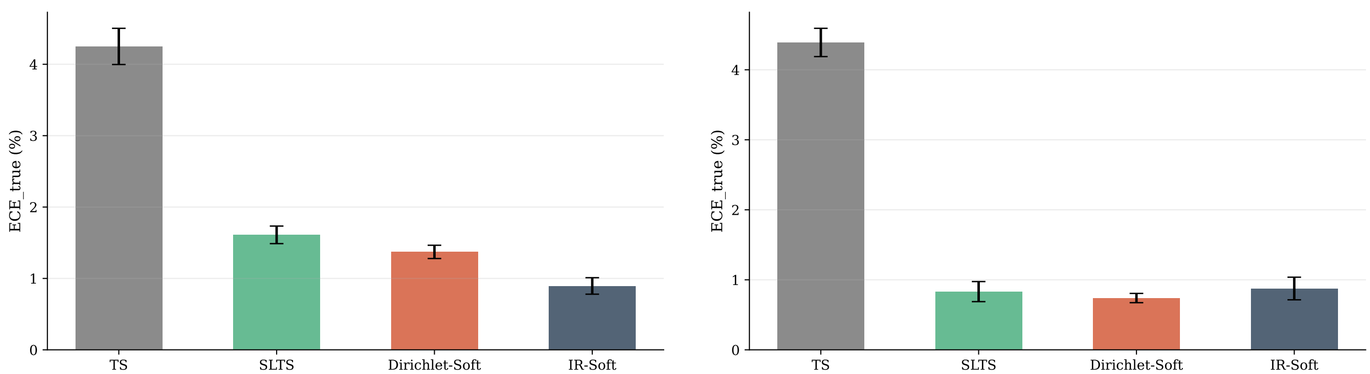

Fig. 15: **Multi-seed bar charts — CIFAR-10H.** Left: ResNet-50. Right: ViT-B/16.

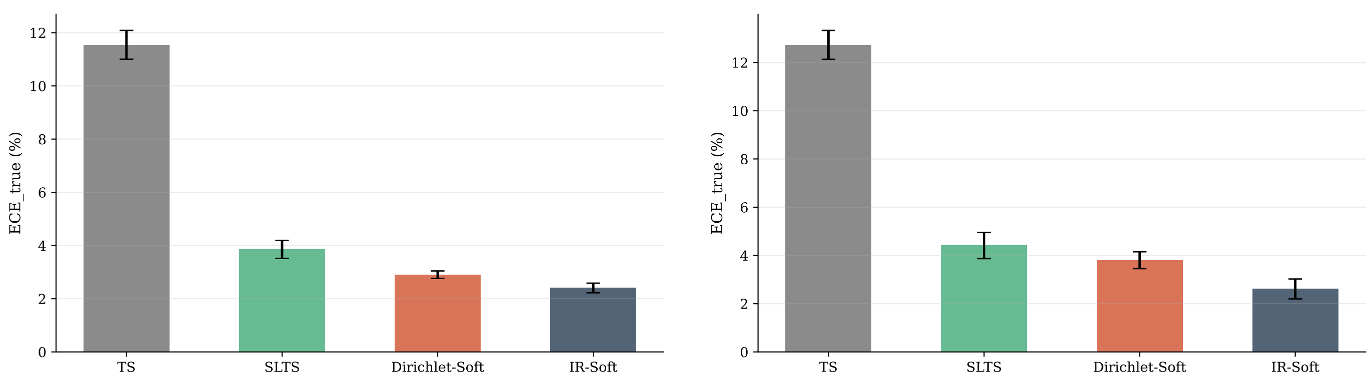

Fig. 16: **Multi-seed bar charts — ChaosNLI.** Left: RoBERTa-Large. Right: DeBERTa-v3-base.

the global mean is a good summary of per-class behaviour for these benchmarks; the slight advantage of CC-LS on DeBERTa-v3 (2.54% vs. 2.65%) is within the single-seed variance reported in Appendix G.

Appendix L
Adaptive Temperature Scaling with Voted Labels (Extended Analysis)

A natural question is whether making the calibration map *adaptive* (predicting a per-instance temperature $T(x_i)$ rather than a global T^*) can close the gap between voted-label calibration and ambiguity-aware calibration, without requiring annotator data. We implement **ATS** [15]: a linear layer maps four logit-derived features $\phi(z_i) = [\max_k z_{ik},\ H(\text{softmax}(z_i))/\log K,\ p_{(1)}-p_{(2)},\ p_{(1)}]$ to $\log T_i$ via $T_i = \text{softplus}(w^\top \phi(z_i)+b)+0.1$, trained with voted-label NLL identical to TS. The four features capture prediction scale, uncertainty, margin, and confidence. We apply ℓ_2 regularisation ($\lambda = 10^{-3}$) to avoid overfitting on small calibration sets.

Finding. ATS fails to improve over TS in all 8 settings and in several cases degrades it (ECE rises from 11.63% to 13.09% on DeBERTa-v3; from 16.59% to 17.53% on ISIC ViT-S/16). The per-instance temperature learned from logit features adapts to *model uncertainty*, not to *annotator disagreement*; these two quantities are loosely correlated at best (cf. Figure 3).

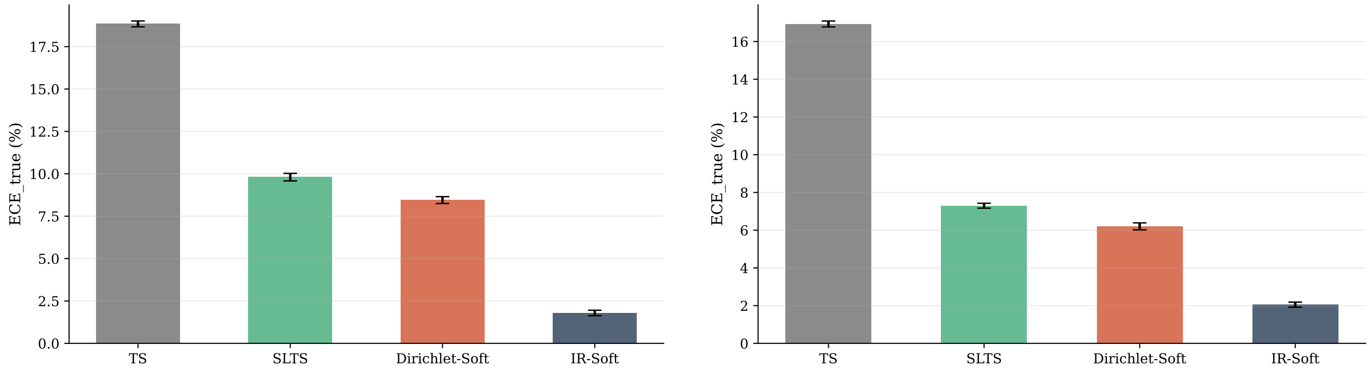

Fig. 17: **Multi-seed bar charts — ISIC 2019.** Left: EfficientNet-B4. Right: ViT-S/16. Seed varies the annotation generation; val/test split is fixed.

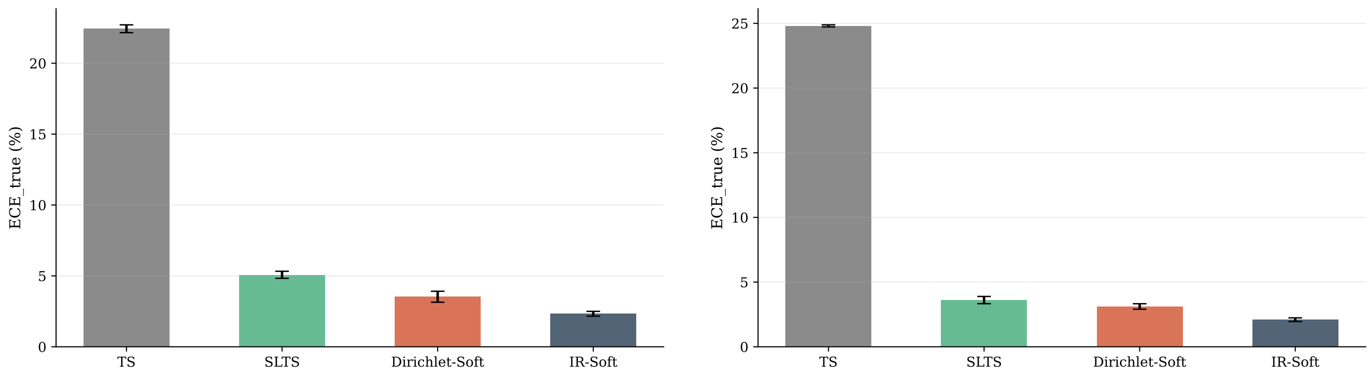

Fig. 18: **Multi-seed bar charts — DermaMNIST.** Left: ResNet-18. Right: ViT-S/16. Seed varies the annotation generation; val/test split is fixed.

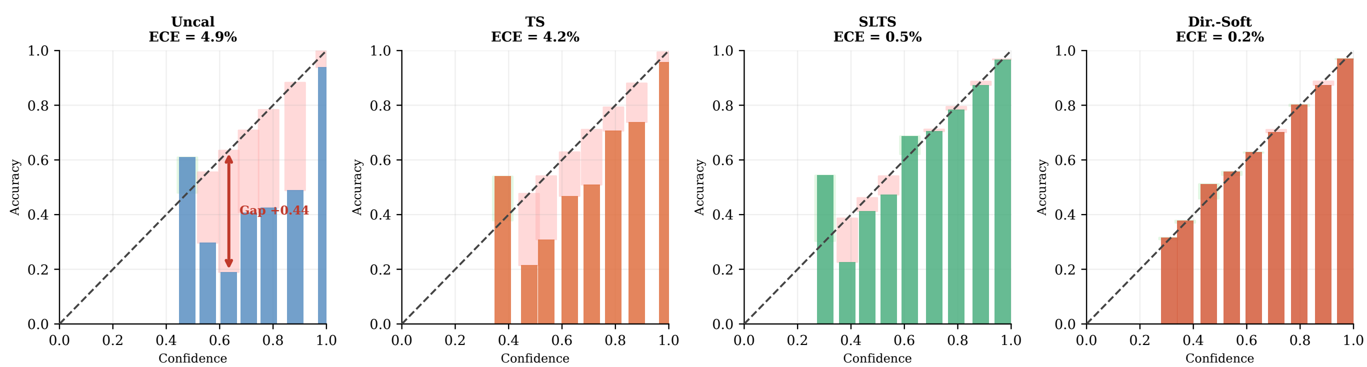

Fig. 19: **Reliability diagrams (summary) — CIFAR-10H ViT-B/16.**

In contrast, LS-TS, which uses the same global-temperature architecture as TS but simply shifts the calibration target toward a smoothed distribution, reduces ECE by 7–78% across all 8 settings without any annotator data. This confirms that the calibration target is the critical design axis: added architectural flexibility without the right supervision signal yields no benefit.

Appendix M
Implementation Details

CIFAR-10H models. *ResNet-50*: ImageNet-1k weights (`ResNet50_Weights.IMAGENET1K_V1`); FC layer replaced with 2048×10 linear; AdamW, lr$= 10^{-4}$, weight decay 10^{-4}, batch 128, cosine annealing, 30 epochs, 224×224. *ViT-B/16*: ImageNet-21k weights; classification head replaced; AdamW, lr$= 5 \times 10^{-5}$, weight decay 10^{-4}, batch 64, cosine annealing, 20 epochs, 224×224.

ChaosNLI models. *RoBERTa-Large*: `roberta-large-mnli` from HuggingFace, pre-fine-tuned on MultiNLI [36]; no additional fine-tuning. *DeBERTa-v3-base*: `cross-encoder/nli-deberta-v3-base` from HuggingFace [37]; no additional fine-tuning. Both models use the combined ChaosNLI SNLI+MNLI subset ($n = 3{,}113$), split 50/50 into calibration and test (stratified, seed 42).

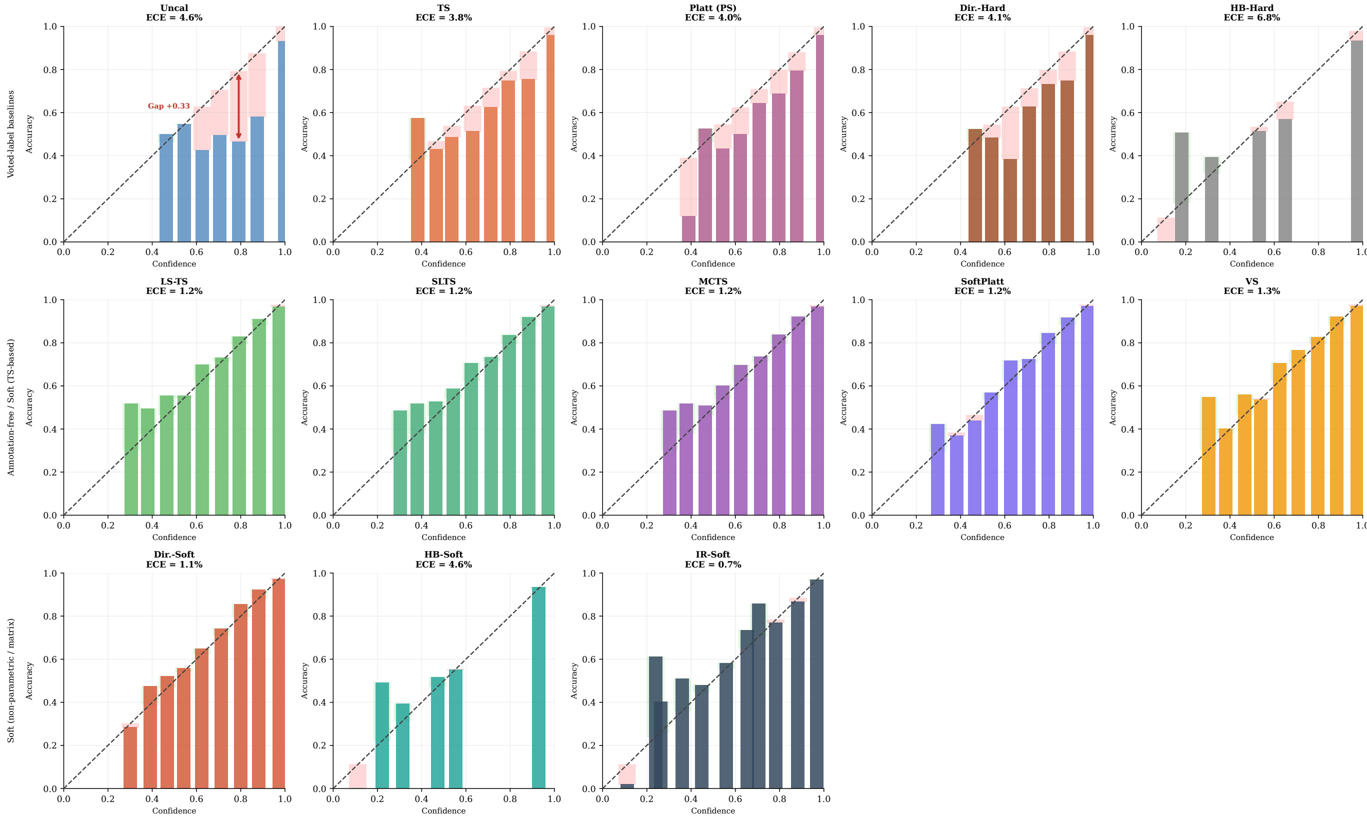

Fig. 20: **Reliability diagrams (all methods) — CIFAR-10H ResNet-50 ($\mathrm{ECE_{true}}$).**

TABLE 8: **ATS vs. TS and LS-TS across all 8 settings (annotation-free methods only) — full results.** $\mathrm{ECE_{true}}$ (%) and Brier. ATS predicts a per-instance temperature from logit features but is trained with the same voted-label NLL as TS. **Bold** = best annotation-free method per column. Same evaluation protocol note as Table 3.

	C10H R50		C10H ViT		NLI RoBERTa		NLI DeBERTa		ISIC ENet		ISIC ViT		Derm R18		Derm ViT	
Method	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br
TS	3.90	.113	4.01	.104	10.55	.537	11.63	.549	18.54	.560	16.59	.617	23.05	.686	24.36	.683
ATS	4.10	.113	4.10	.105	11.36	.538	13.09	.552	18.83	.560	17.53	.631	22.82	.687	25.35	.686
LS-TS	**1.49**	**.111**	**1.62**	**.102**	**4.16**	**.522**	**2.65**	**.529**	**9.34**	**.528**	**15.49**	**.619**	**5.05**	**.630**	**7.36**	**.615**

ISIC 2019 models. *EfficientNet-B4*: ImageNet-1k weights; stratified 70/15/15 split; class-weighted cross-entropy (NV: $\approx$67% of training images); AdamW, lr$= 3 \times 10^{-4}$, weight decay 10^{-4}, 2-epoch linear warm-up + cosine decay, 20 epochs, 380×380. *ViT-S/16*: ImageNet-21k weights; same split and loss weighting; AdamW, lr$= 10^{-4}$, weight decay 10^{-4}, 2-epoch warm-up + cosine decay, 20 epochs, 224×224.

DermaMNIST models. *ResNet-18*: ImageNet-1k weights, class-weighted cross-entropy, AdamW, lr$= 10^{-4}$, 30 epochs. *ViT-S/16*: ImageNet-21k weights, same training protocol, 224×224.

Calibration optimisers. TS, SLTS: LBFGS, lr$= 0.1$, 500 iterations, tolerance 10^{-9} (gradient), 10^{-11} (loss). Platt/VS: Adam, lr$= 0.01/0.05$, weight decay 10^{-4}, 2000 steps.

ECE bins. $B = 15$ equal-width bins in $[0, 1]$; bins with fewer than 1 example are excluded.

Compute. CIFAR-10H fine-tuning: $\approx$20 min on one NVIDIA A100. All calibration methods: < 60 s on CPU after logit caching.

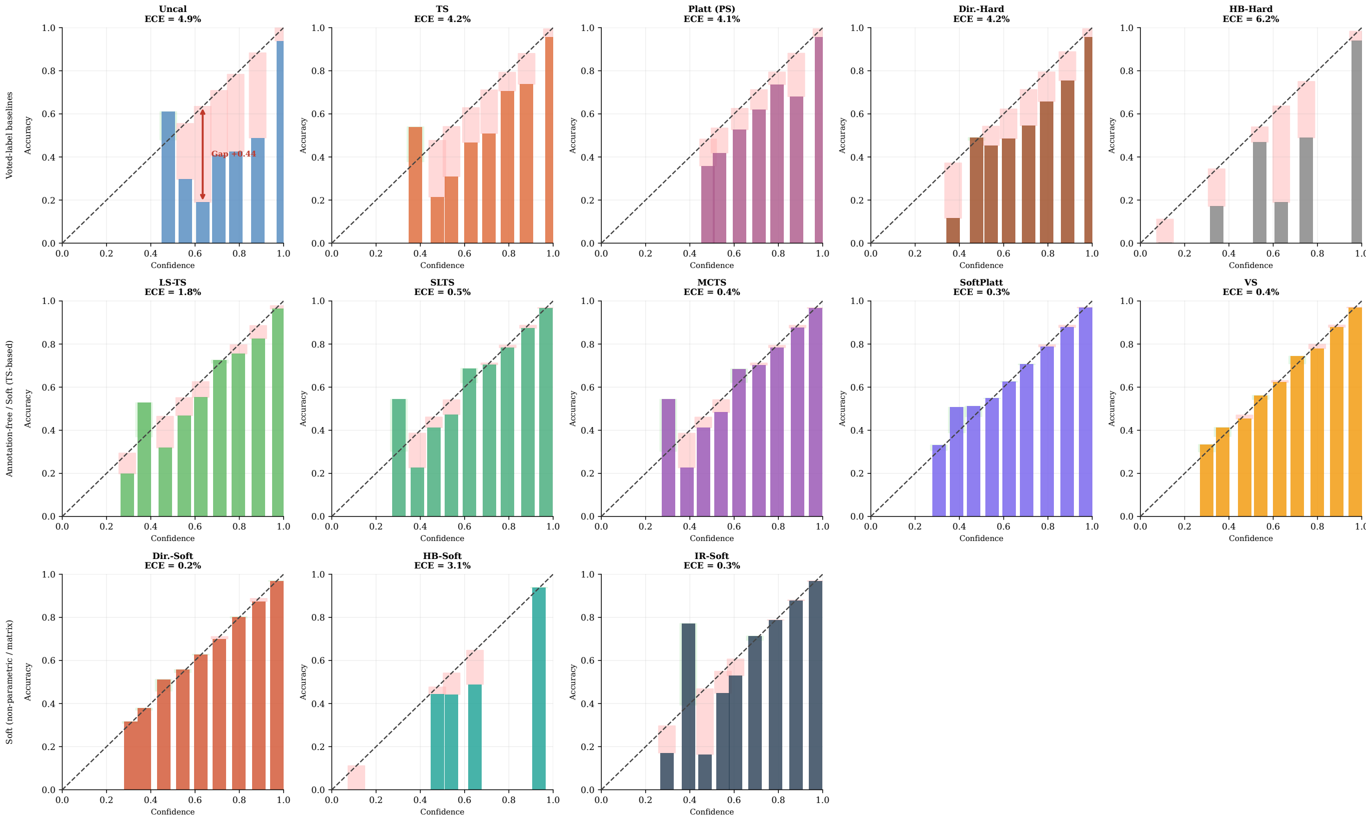

Fig. 21: **Reliability diagrams (all methods) — CIFAR-10H ViT-B/16 (ECE$_{\text{true}}$).**

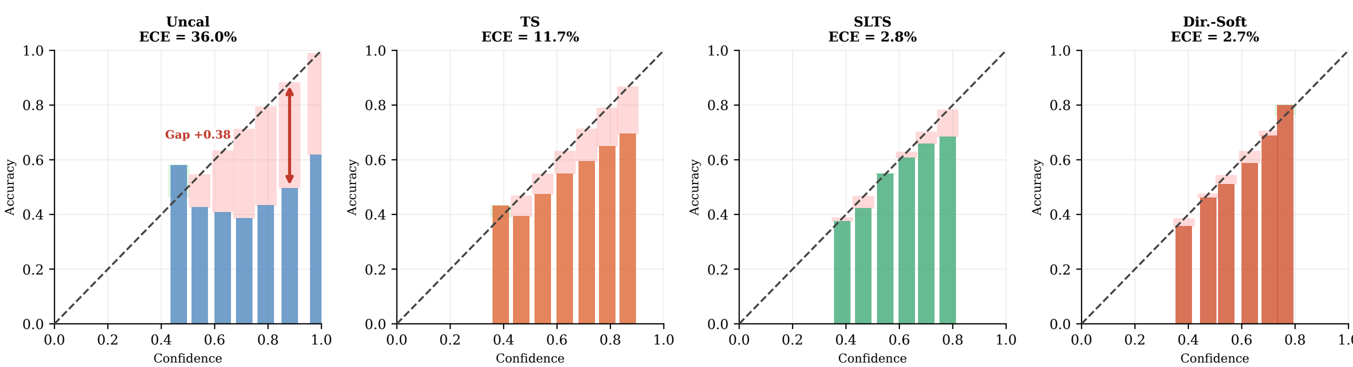

Fig. 22: **Reliability diagrams (summary) — ChaosNLI DeBERTa-v3-base.**

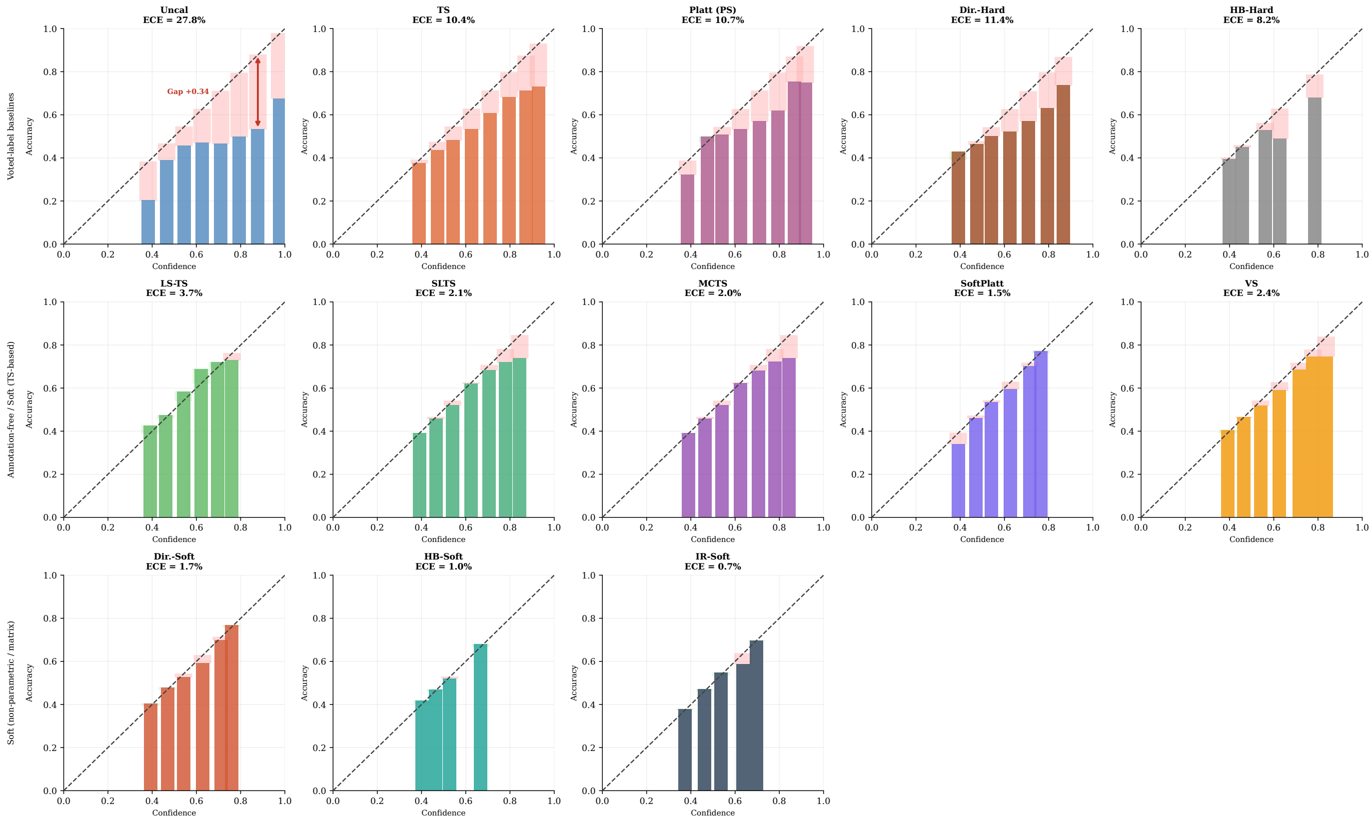

Fig. 23: **Reliability diagrams (all methods) — ChaosNLI RoBERTa-Large ($\mathbf{ECE_{true}}$).** NLI models are severely overconfident before calibration; ambiguity-aware methods substantially correct this.

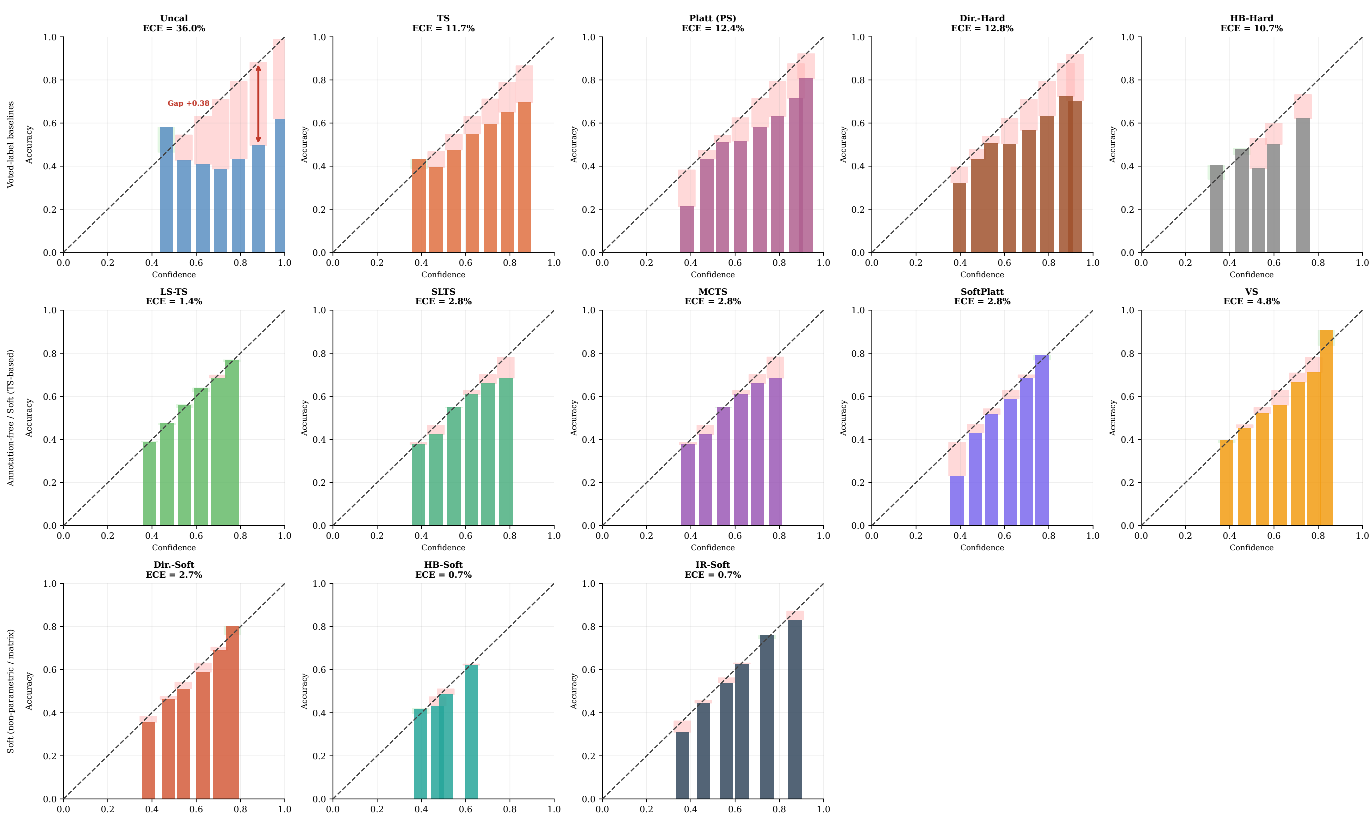

Fig. 24: **Reliability diagrams (all methods) — ChaosNLI DeBERTa-v3-base ($\mathbf{ECE_{true}}$).**

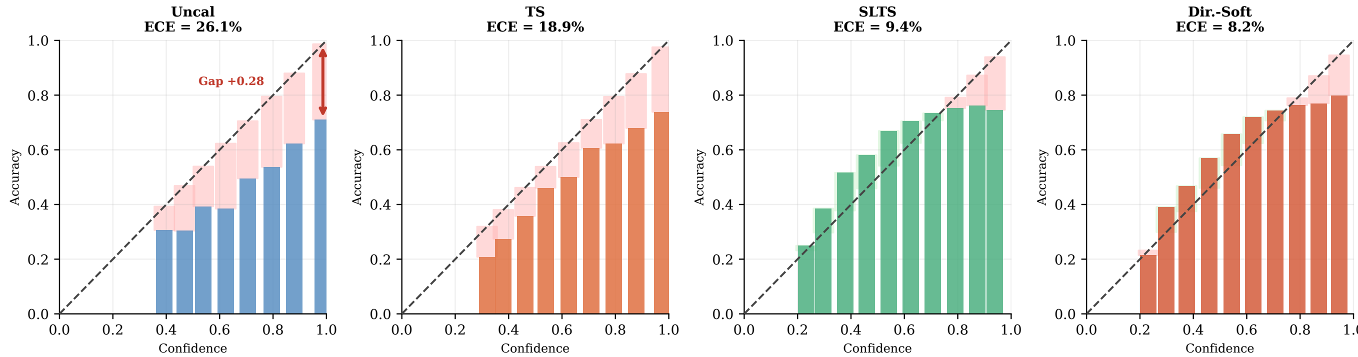

Fig. 25: **Reliability diagrams (summary) — ISIC 2019 EfficientNet-B4.**

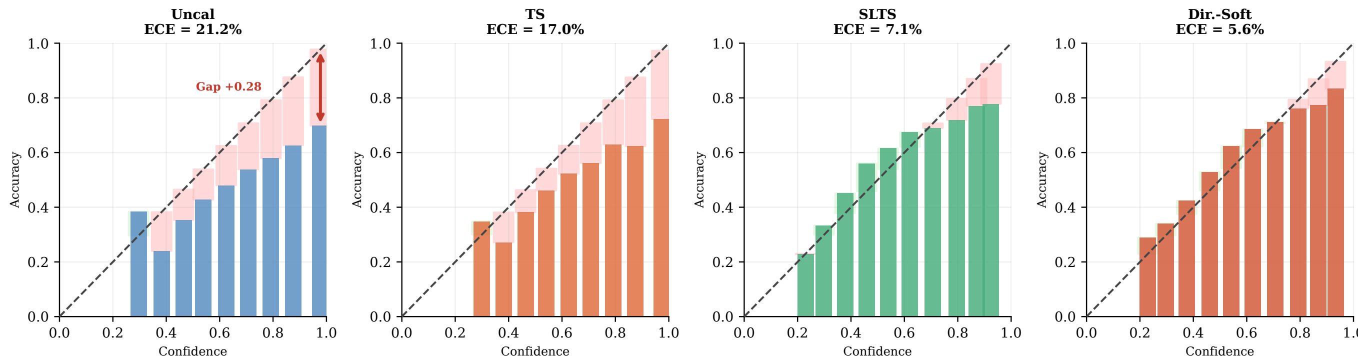

Fig. 26: **Reliability diagrams (summary) — ISIC 2019 ViT-S/16.**

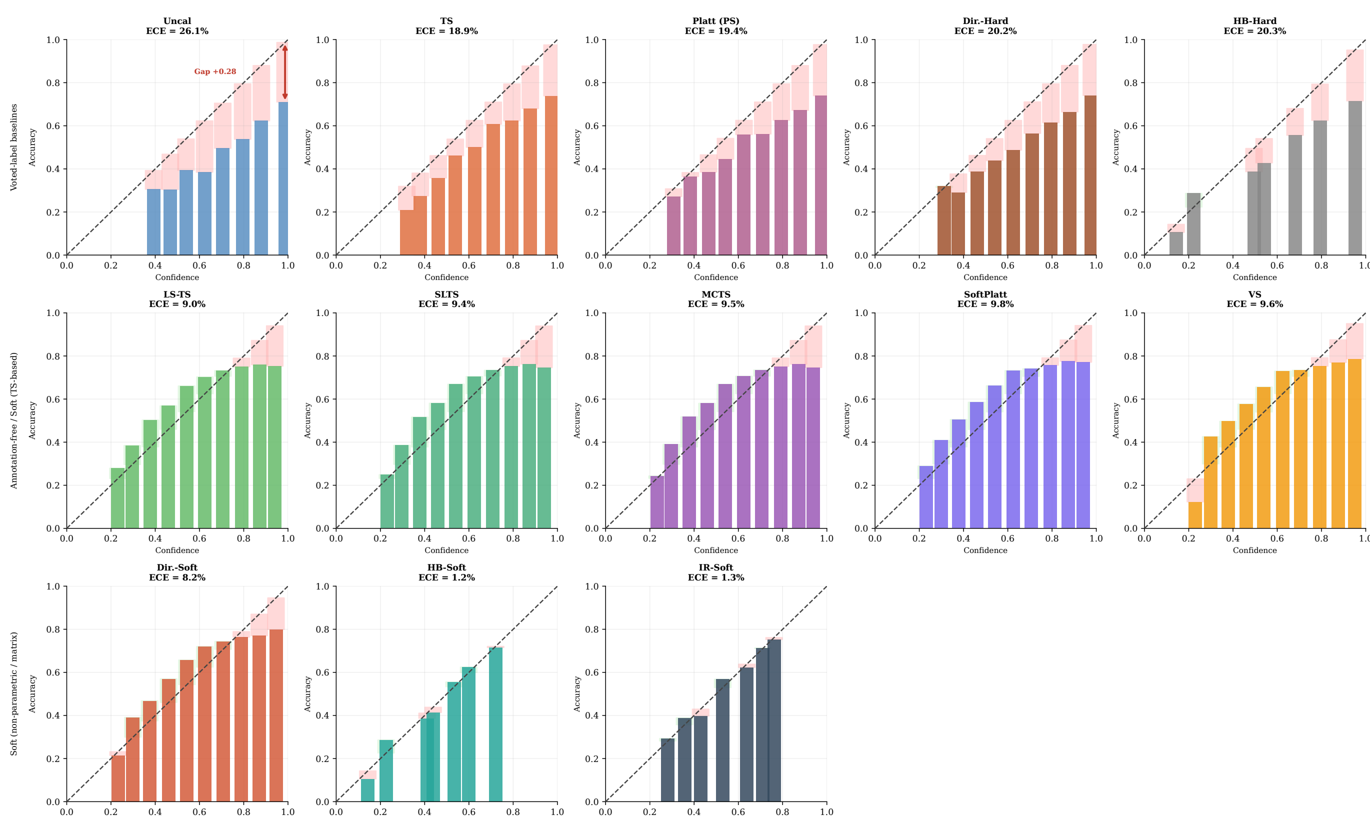

Fig. 27: **Reliability diagrams (all methods) — ISIC 2019 EfficientNet-B4 (ECE_{true}).**

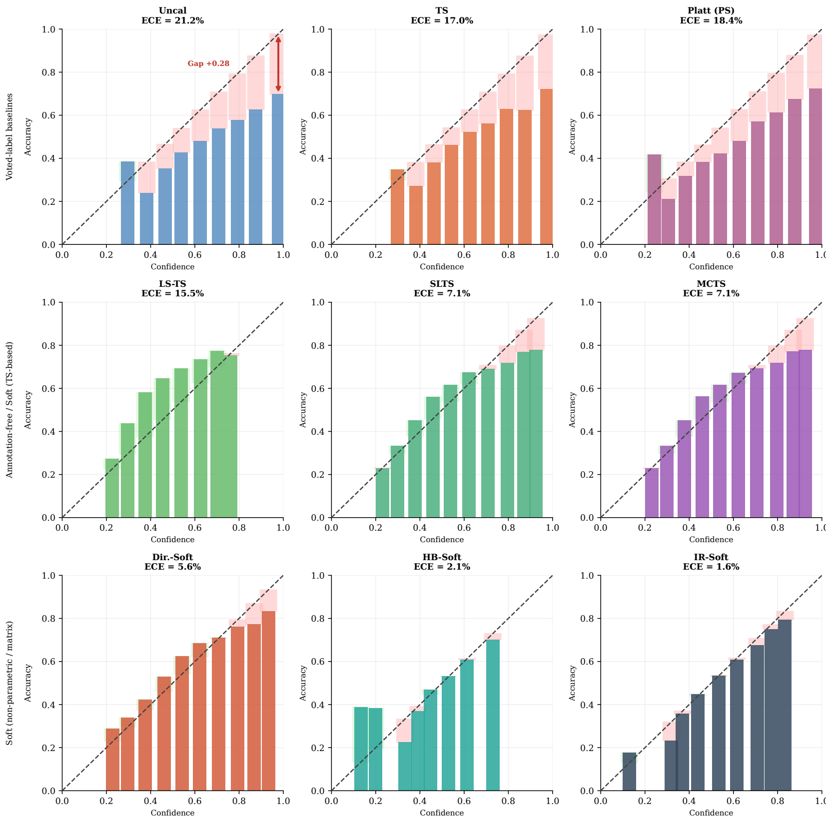

Fig. 28: **Reliability diagrams (all methods) — ISIC 2019 ViT-S/16 (ECE_{true}).**

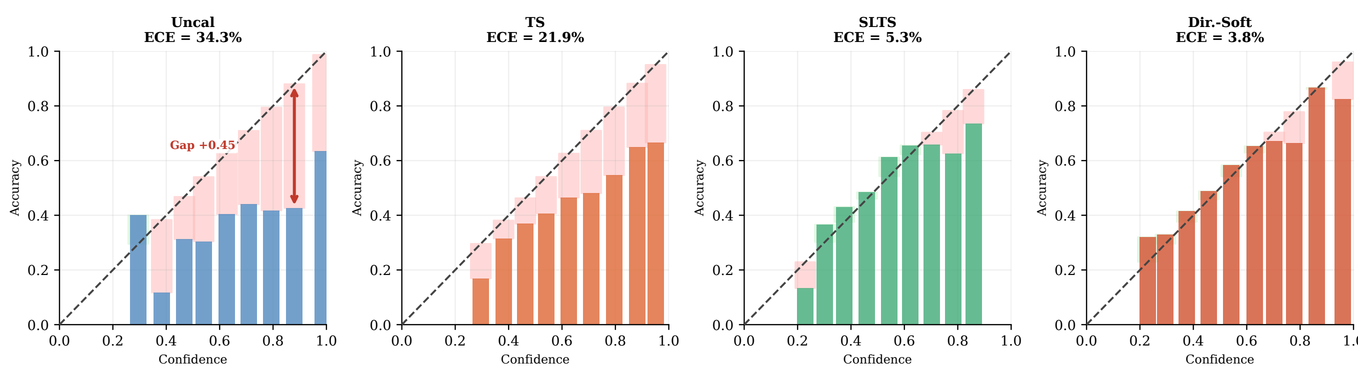

Fig. 29: **Reliability diagrams (summary) — DermaMNIST ResNet-18.**

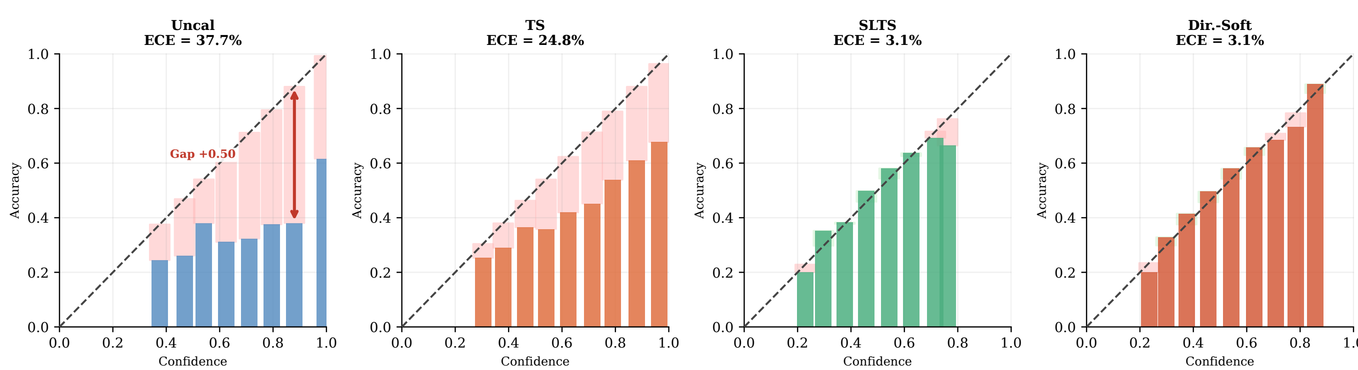

Fig. 30: **Reliability diagrams (summary) — DermaMNIST ViT-S/16.**

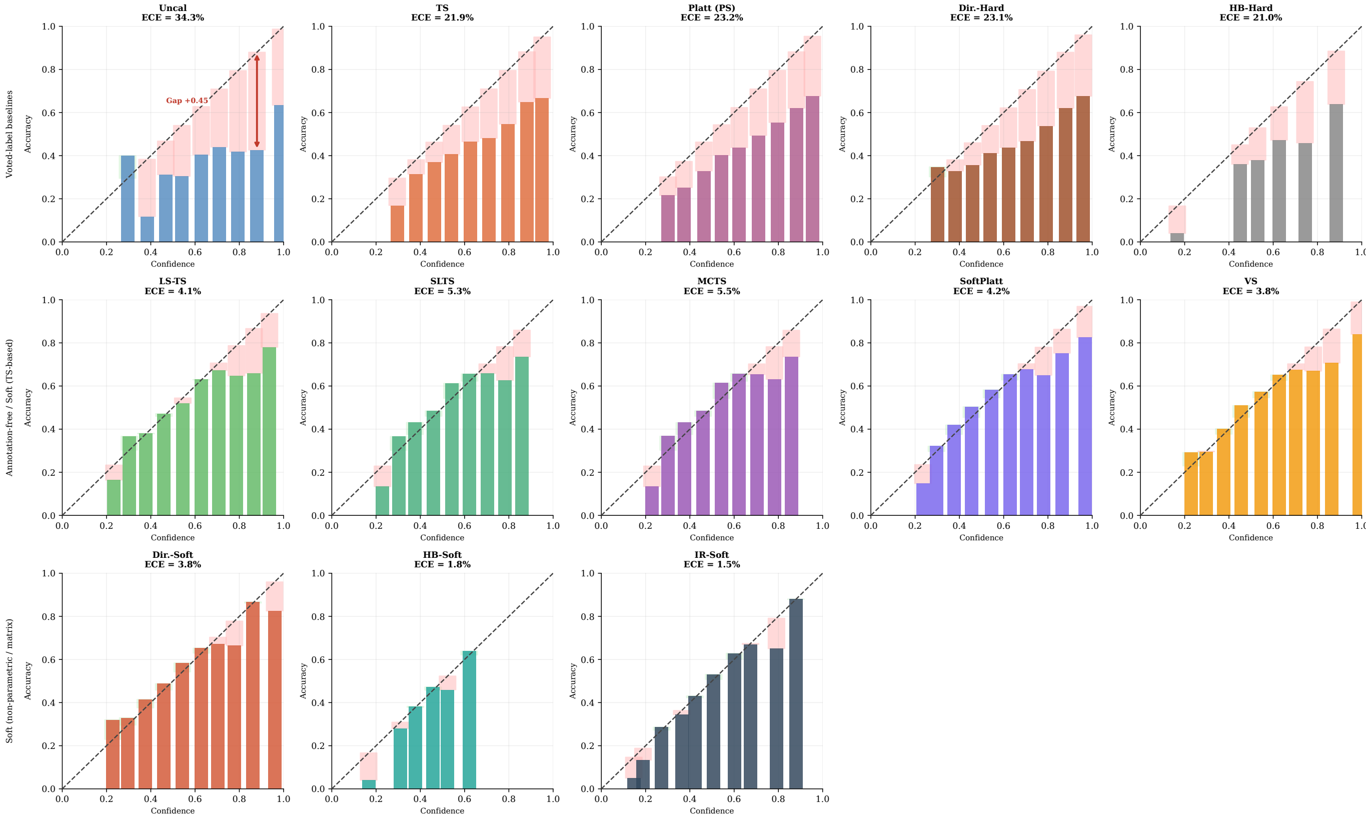

Fig. 31: **Reliability diagrams (all methods) — DermaMNIST ResNet-18 ($\mathrm{ECE}_{\mathrm{true}}$).**

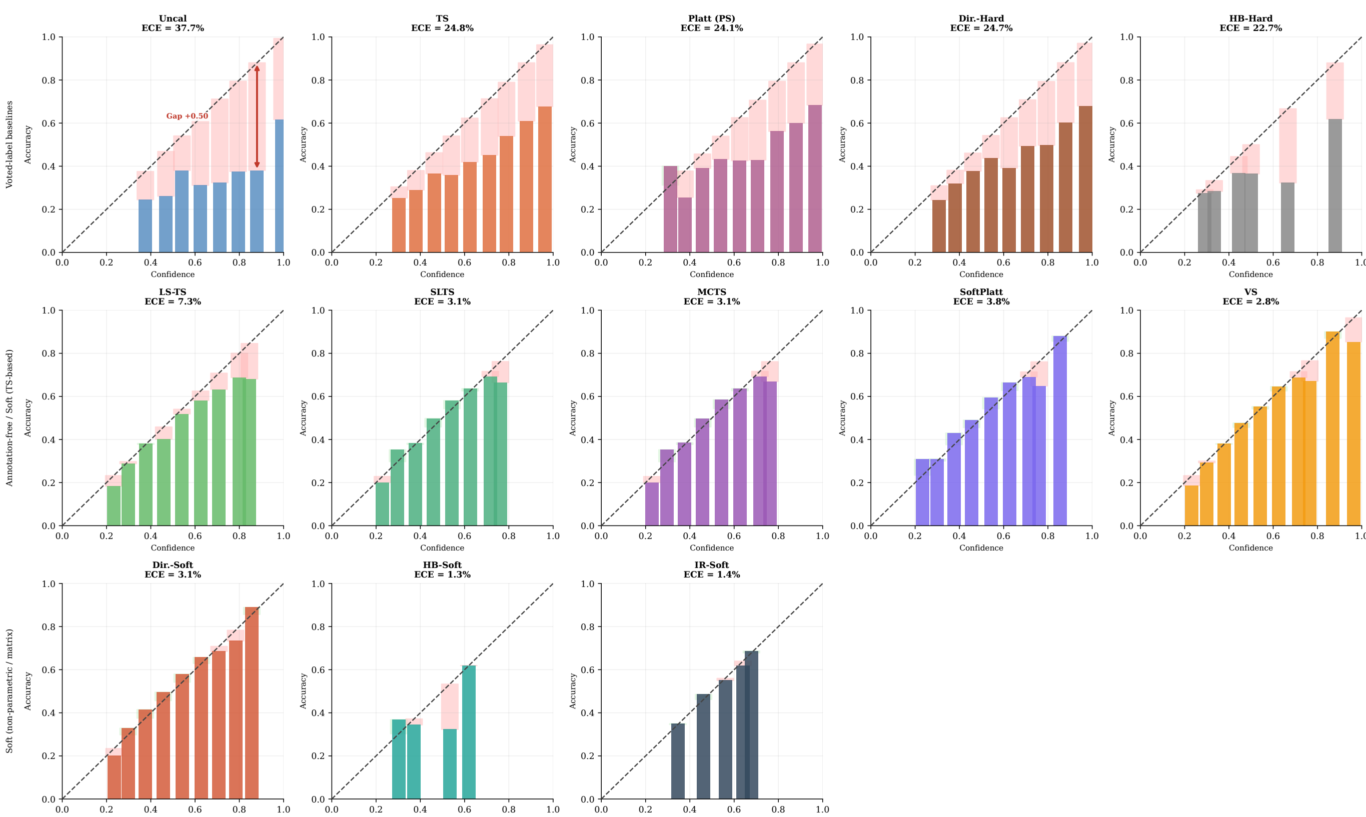

Fig. 32: **Reliability diagrams (all methods) — DermaMNIST ViT-S/16 ($\mathrm{ECE}_{\mathrm{true}}$).**

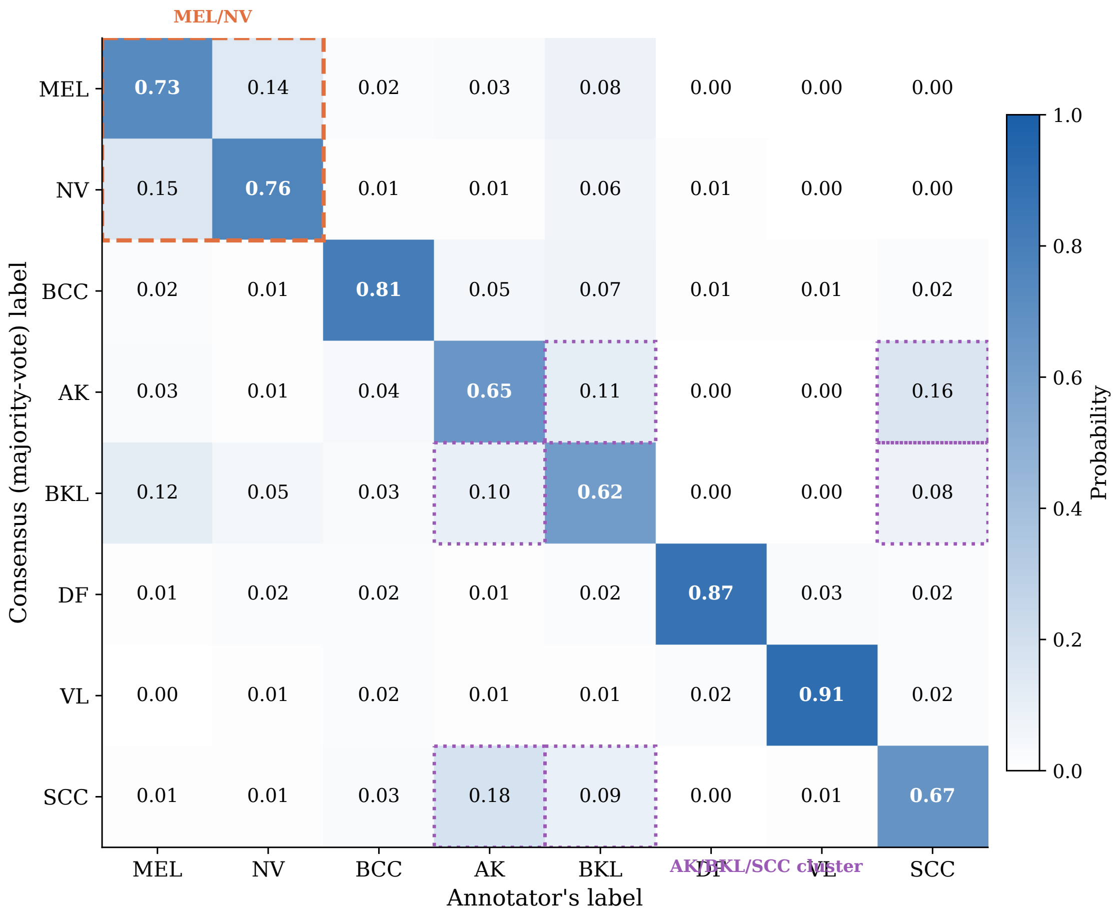

Fig. 33: **Inter-reader confusion matrix for ISIC 2019.** Entry C_{ij} is the probability that a dermatologist labels an image as class j given the consensus label is i. Diagonal entries (bold) are the per-condition agreement rates, calibrated to match Liu et al. [41] Table 2. Orange dashed box: MEL/NV most-confused pair. Purple dotted highlights: AK/BKL/SCC high-confusion cluster. Mean diagonal agreement $\bar{C}_{ii} = 75\%$.